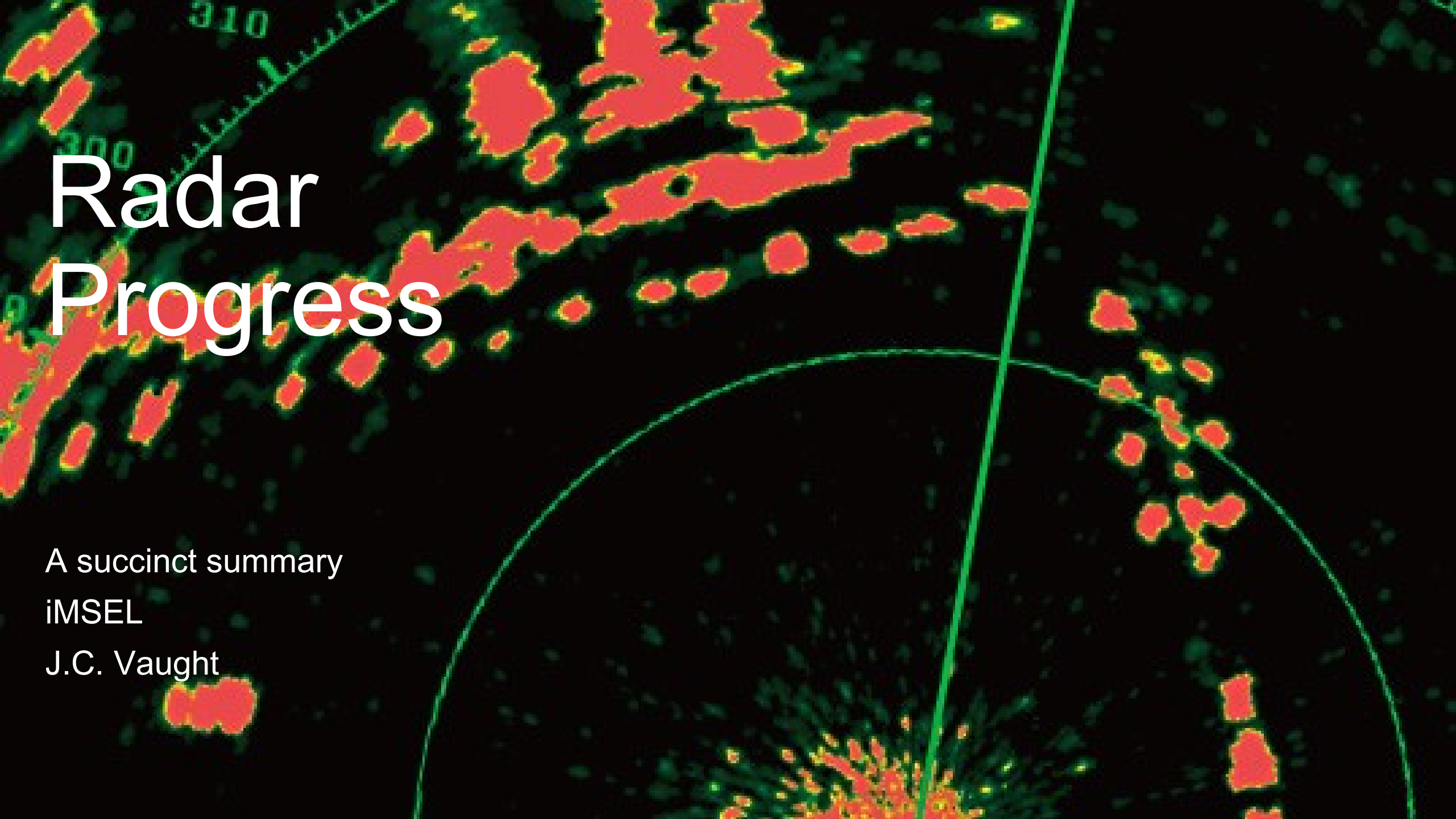




Radar Progress

A succinct summary
iMSEL
J.C. Vaught

Overview



Part 1: Literature Review



Part 2: ARPA vs mARPA



Part 3: Work Progress

PART 1: Literature Review – Overview

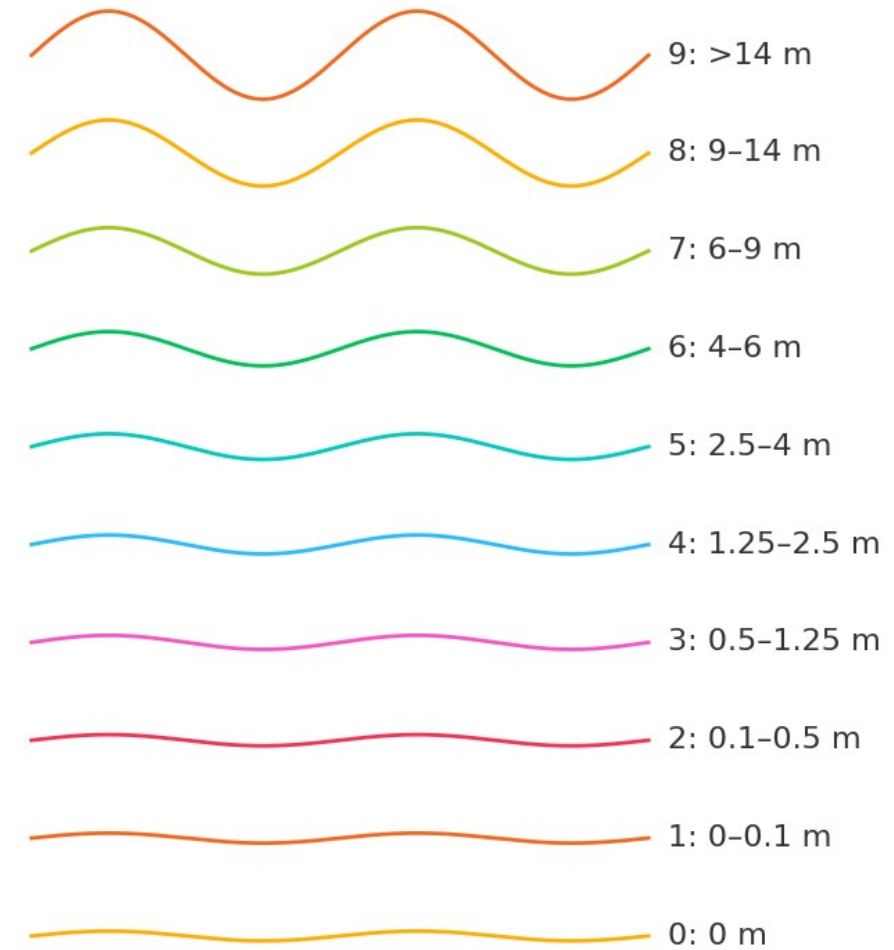
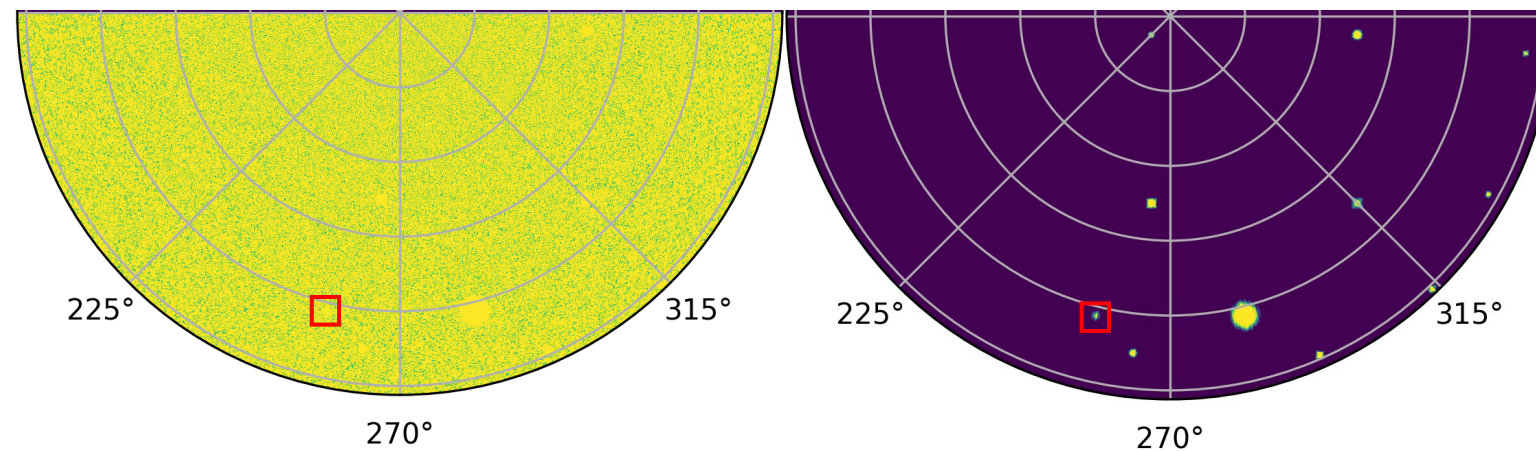
- Looked at 142 Papers/Book chapters spanning '49 to 2025
- Retrieved from IEEE, Elsevier, arXiv – (some pirated)
- Subject = small object detection/classification with Radar
- 3 primary themes: Radar, Vision, and Fusion-Sensor Systems

* *NOT a 'true' literature review.*



Literature Review – Detection Challenges

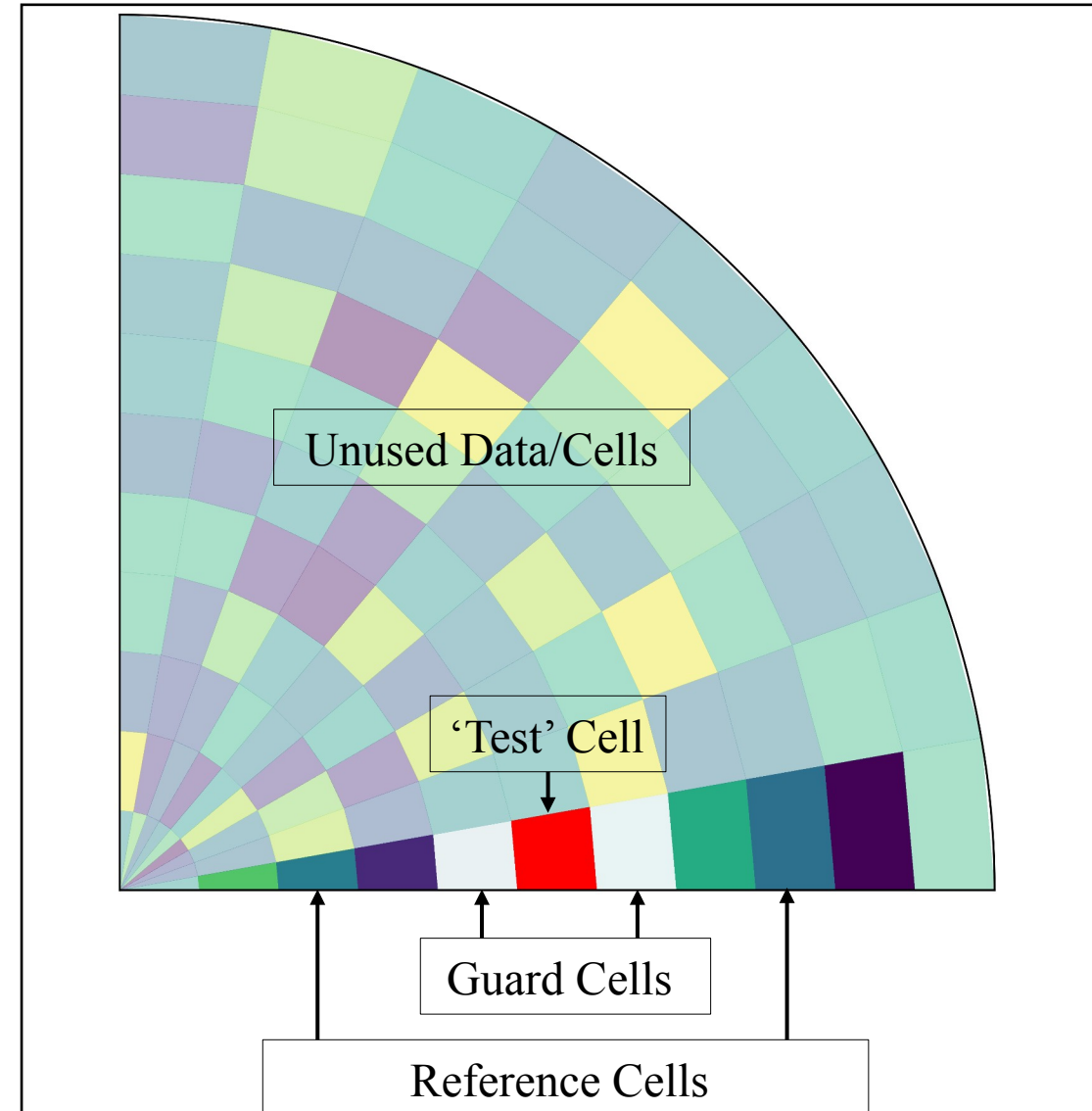
- Low RCS objects & Sea Clutter bury signal [1,2]
- Vision fails in fog/night [23]
- < 0.3 Pd for swimmers in Sea-State 4 [19]



Douglas Sea States

Literature Review – Radar-Centric Solutions

- CFAR variants dominate(30+% of papers)
- Doppler/MTI for moving targets[63]
- Millimetre-wave FMCW rising after 2019 [89]
- Polarimetry & adaptive waveforms emerging



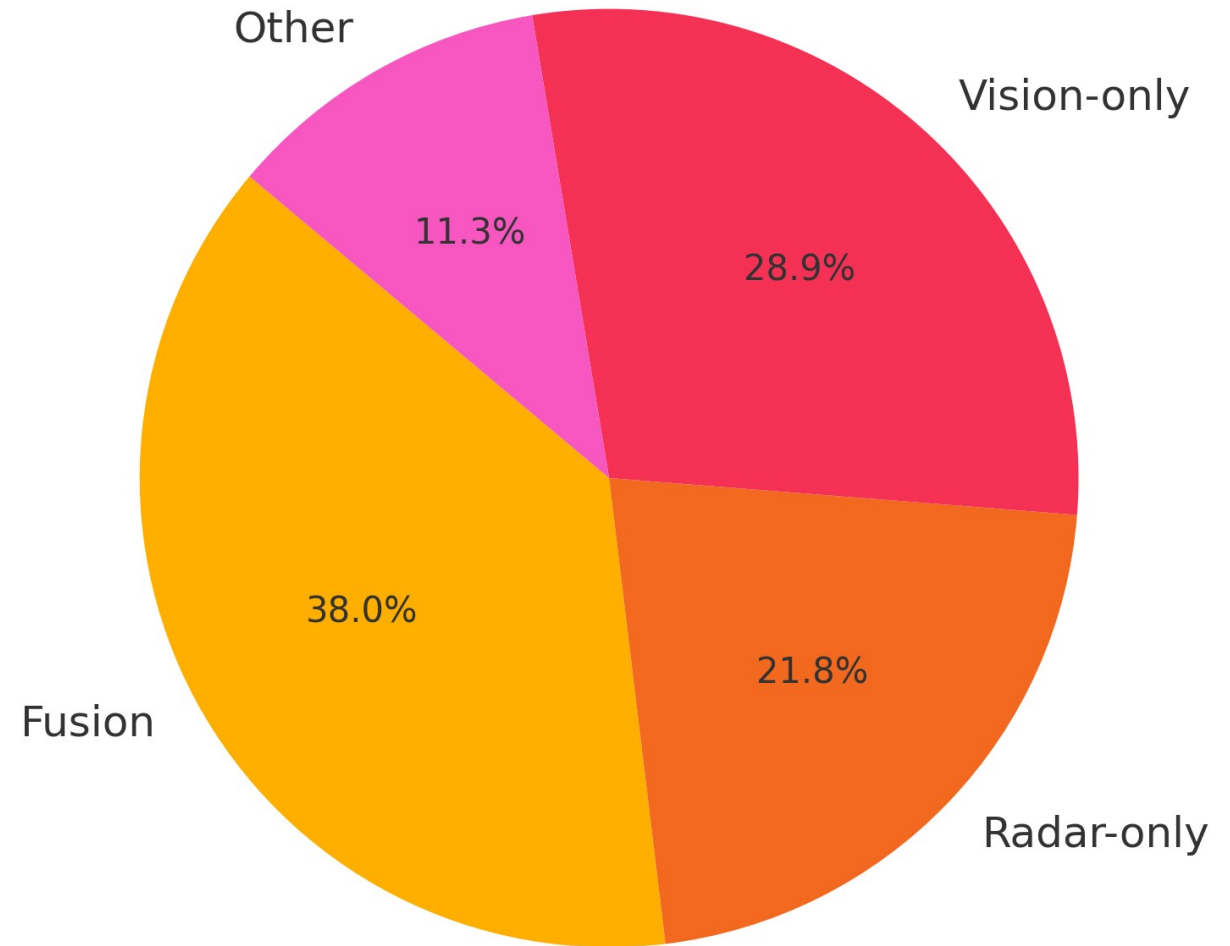
Literature Review – Vision-Only Solutions

- Deep CNNs (YOLO, Faster R-CNN) on ship datasets [100]
- Synthetic augmentation adds $\approx 5\%$ mAP
- Little data on small objects(<32px) in marine contexts
- Lots of augmented solutions to help with lack of data in ocean



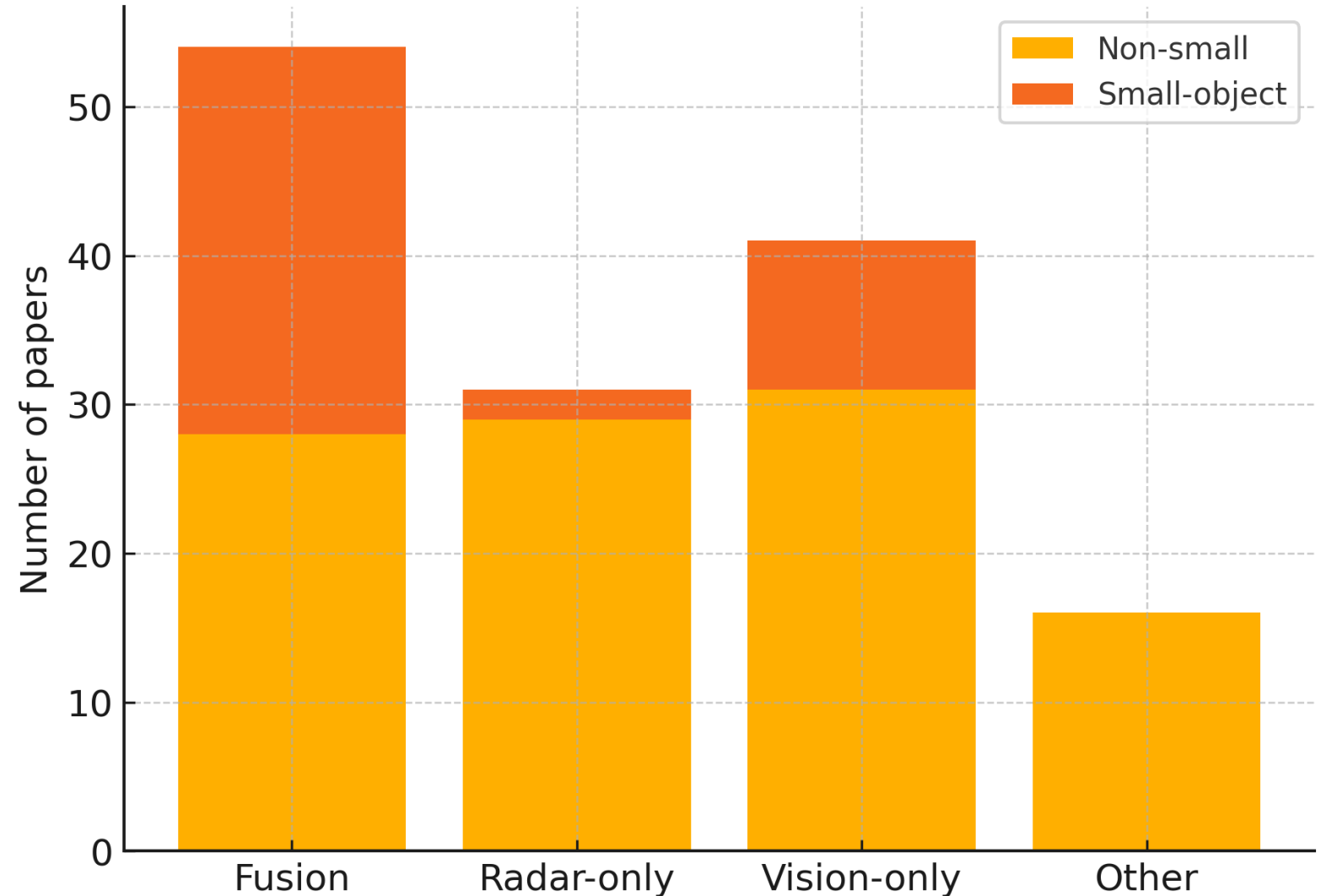
Literature Review – Sensor-Fusion Studies

- 54 papers (38% of corpus)
- Radar-cued ROI → camera classifier [RRPN 2019]
- Calibration error ranges 2.6–18 px
- Few maritime datasets (PoLaRIS 2024)
 - This is increasing ^^

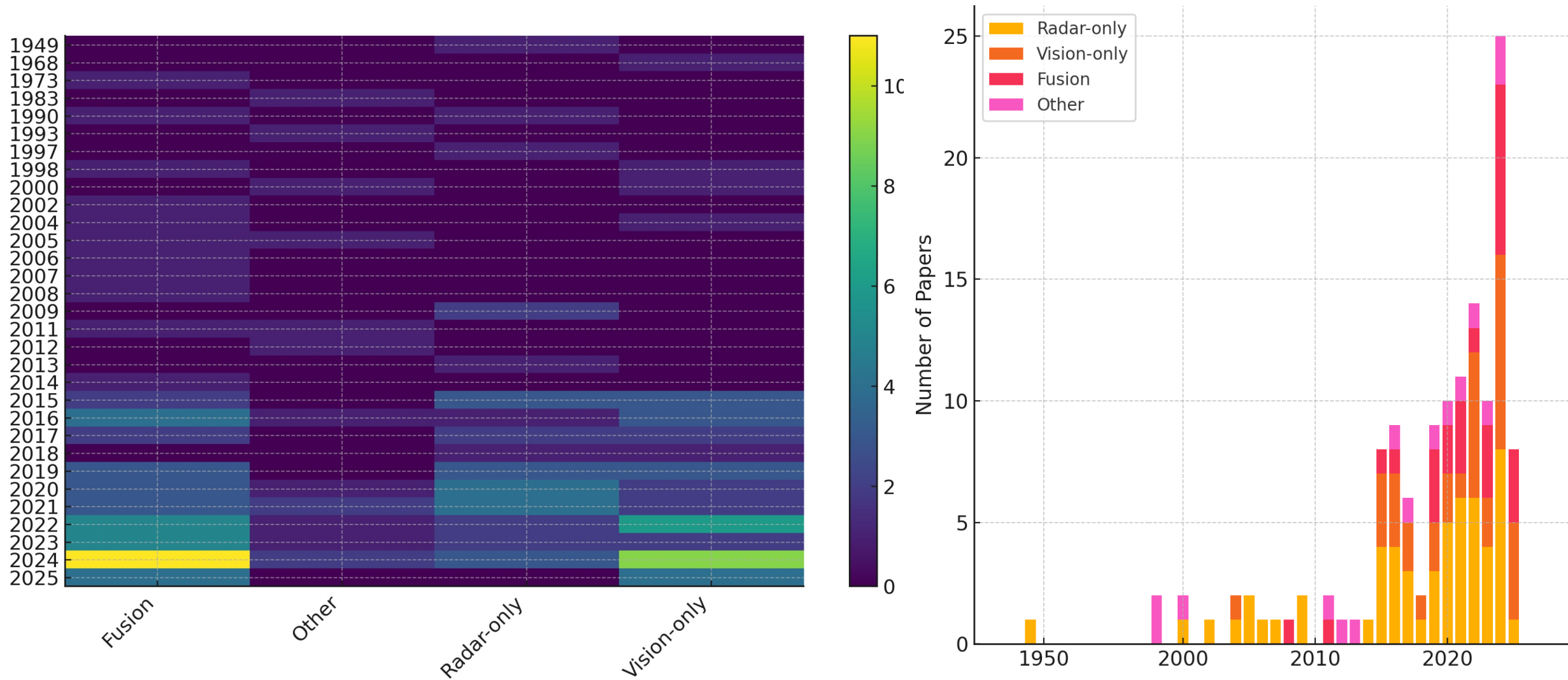


Literature Review – Small-Object Work

- Only 6 % of **radar** papers focus on tiny targets
- **Vision**: 25 %
- **Fusion**: slight minority are small-object

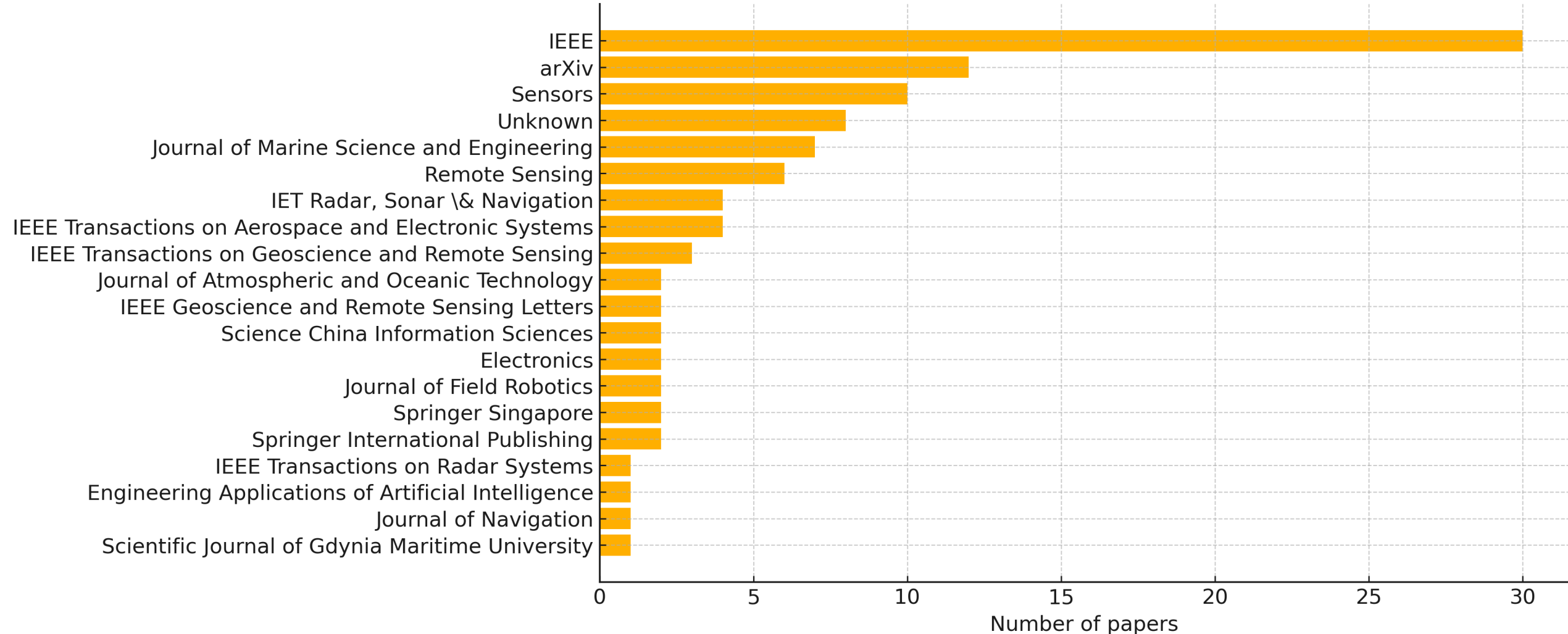


Literature Review – Time Trends

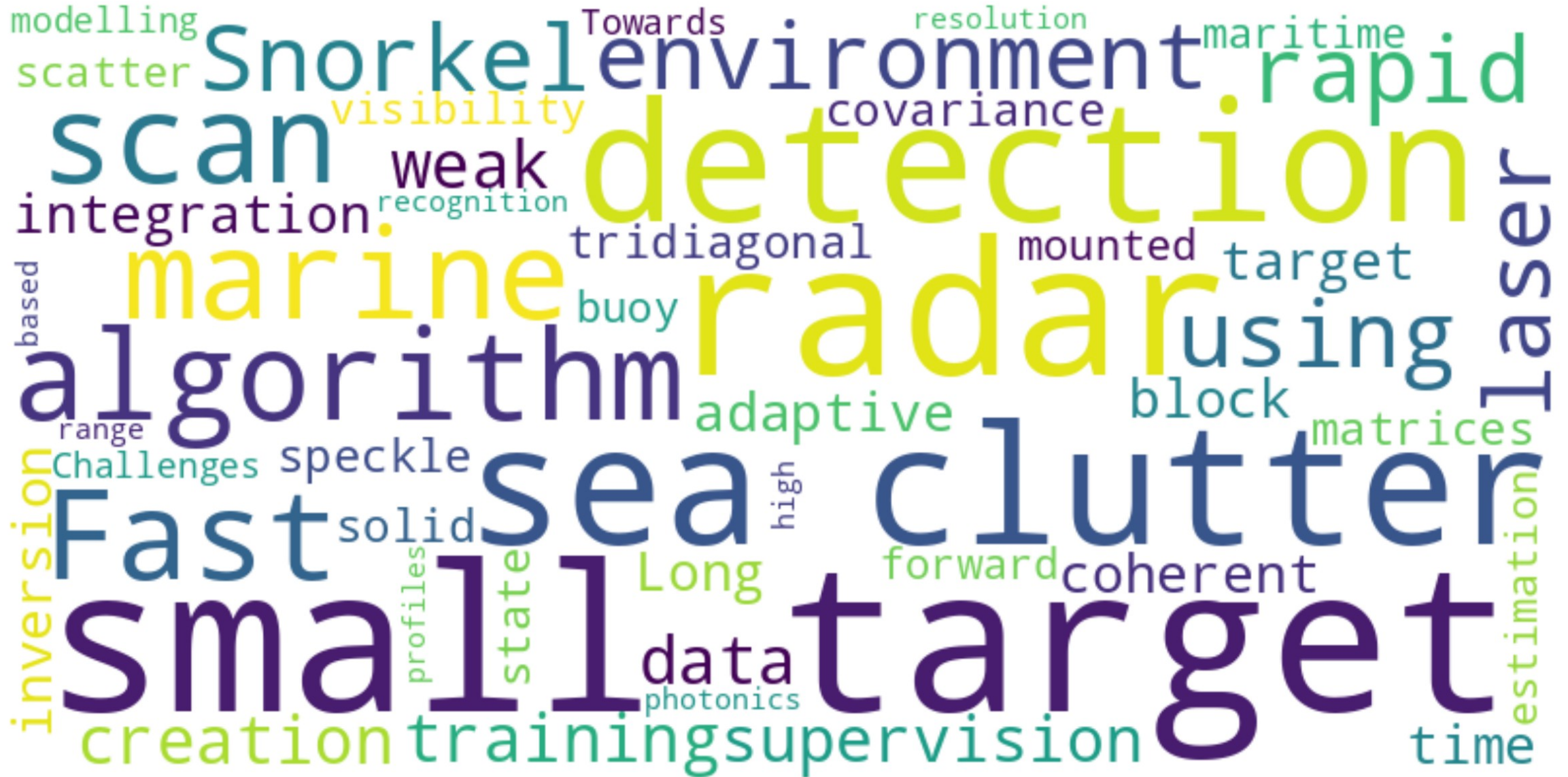


Literature Review – Top Published Venues

Top-20 publication venues (142-paper corpus)



Literature Review – ‘Buzzwords’



Literature Review: Takeaways / Synthesis

- **Radar-Only Detection Challenges**

- Very low RCS ($<0.1 \text{ m}^2$) targets buried in dynamic sea clutter
- High false-alarm rates in rough seas

- **Vision-Only Data Scarcity**

- Public maritime datasets small (median ~ 600 frames)
- Limited diversity hinders deep-learning generalization

- **Radar-Guided Approaches**

- RRPN boosts small-object recall by $\sim 15\%$
- Fusion methods yield $\geq 10\%$ gain in Pd

- **Remaining Challenges**

- Calibration / projection misalignments ($1^\circ = \text{meters of error}$)
- Radar false positives from sea spikes

PART 2: ARPA/mARPA – ARPA Core Functions

- Detects / acquires targets (auto or manual).
- Updates tracks every antenna rotation (≤ 2.5 s @ 48 RPM).
- Draws *true* or *relative* motion vectors for each target.
- Calculates CPA & TCPA continuously; triggers audible/visual alarms when thresholds breached.
- Maintains & displays up to 100 tracks on modern sets; target list includes ID, bearing, range, COG, SOG, CPA, TCPA.



ARPA/ mARPA – ARPA requirements



Gyro-compass or equivalent heading sensor ($\leq 0.5^\circ$ error).



Speed input (log or GPS) for *true-motion* vector maths.



Processor capable of per-sweep target correlation & Kalman filtering at 24-48 RPM.



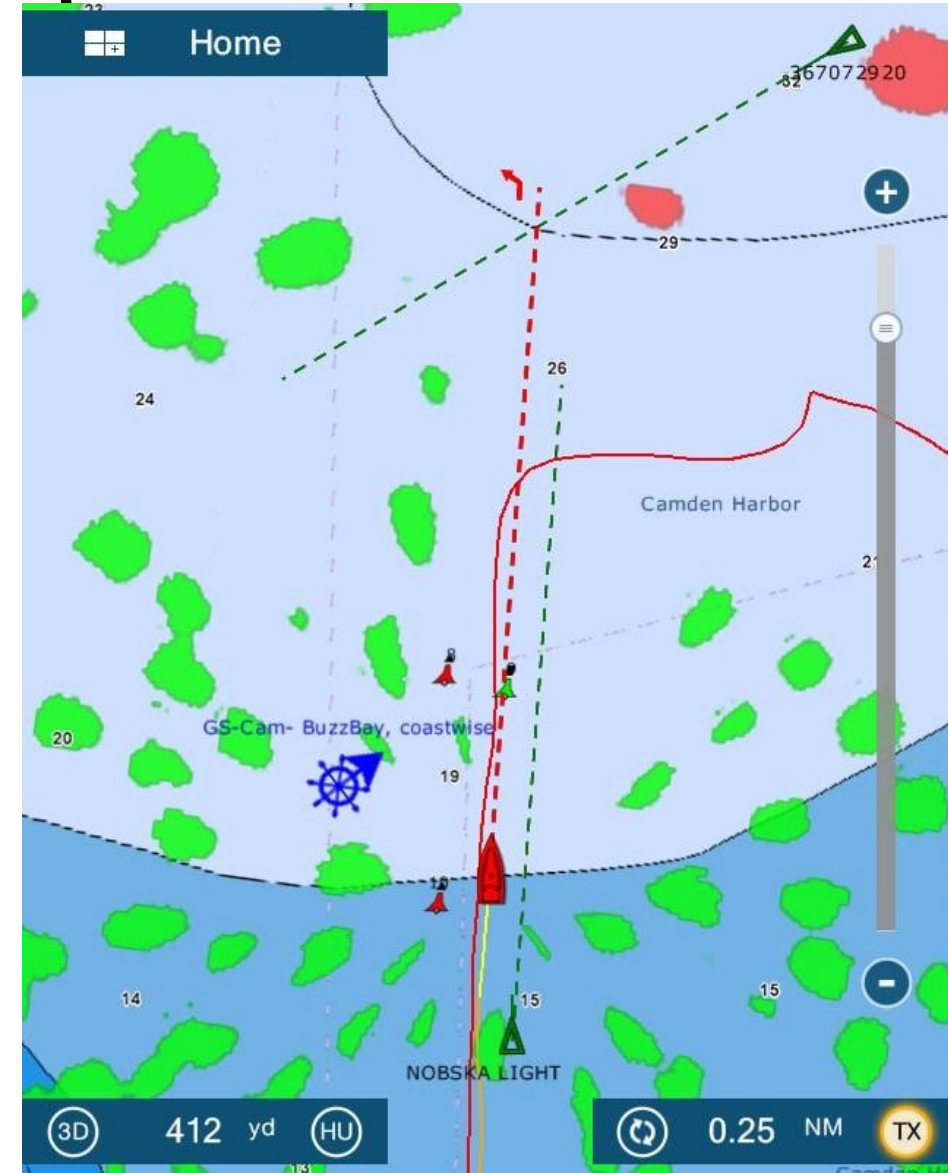
High-resolution radar front-end (pulse-compression, narrow beam) so close echoes are separable.



Network path to the bridge display for presentation & alarms.

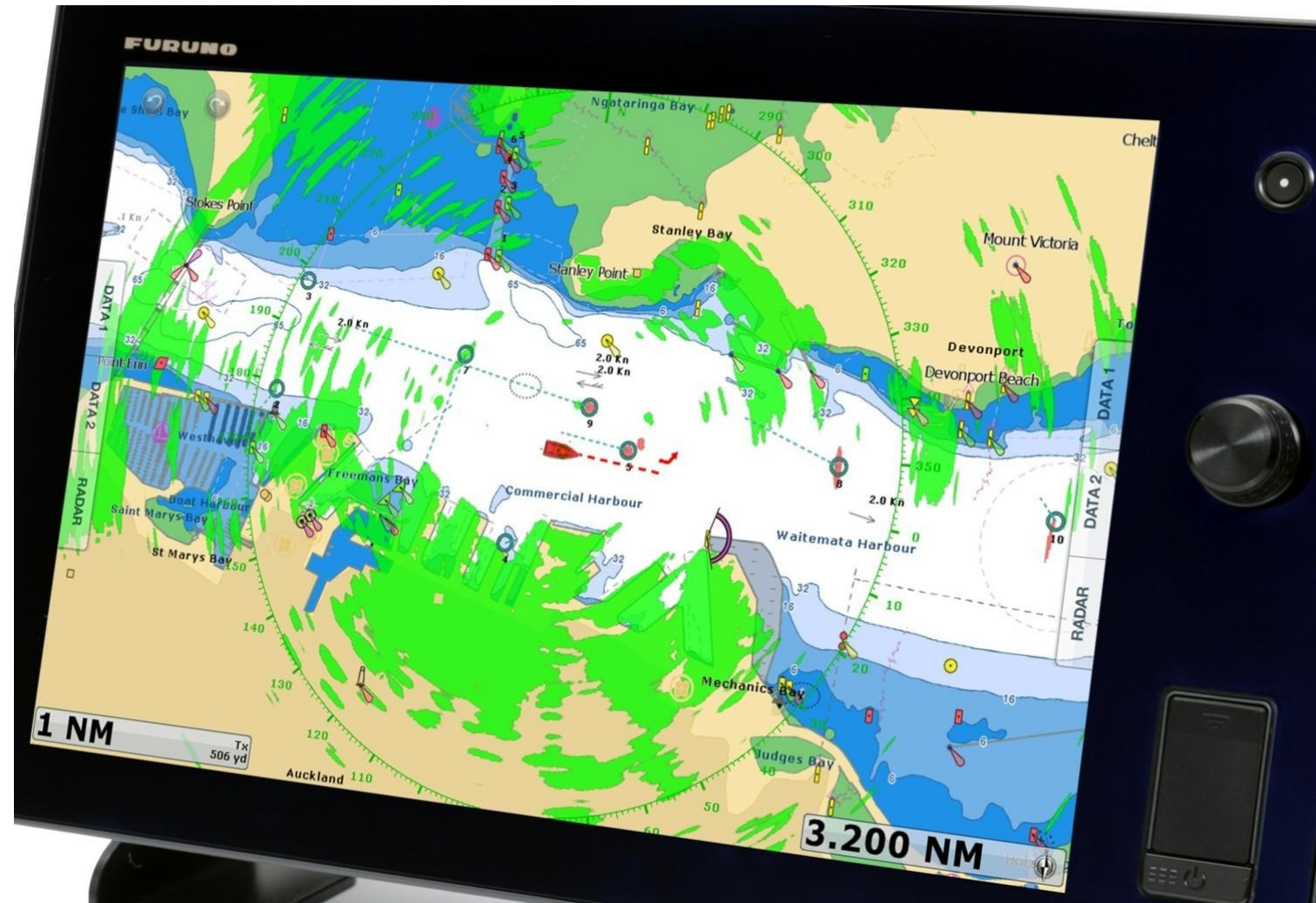
ARPA/mARPA – Target Acquisition

- **Automatic:** radar adopts rule set (e.g., Doppler closure > 3 kn, inside guard ring) and self-plots.
- **Manual:** operator selects an echo; needed for stationary or receding targets.
- Most small-craft radars: 10-30 manual tracks (mARPA). Full ARPA (e.g., Furuno NXT) ≈ 100 auto + manual tracks.
- Generally run auto mode underway; fall back to manual/guard mode only in dense clutter.



ARPA/mARPA – What it looks like

- Closing targets rendered *red* (≥ 3 knots); stationary / opening echoes in *green*.
- White vector shows where each target will be in user-set time (e.g., 1 min).
- On-screen CPA symbol marks exact crossing point; data-box gives range, bearing, CPA, TCPA.
- Operator sees which vessel is dangerous and how soon it'll arrive even in fog/night.



ARPA/ mARPA – What is m(ini)ARPA



“Mini-ARPA” on small-craft radars.



User must manually tag each target.



Typically 10–30 targets; slower update cadence.



No automatic hazard selection.



Limited target count & range; may drop tracks after course changes.



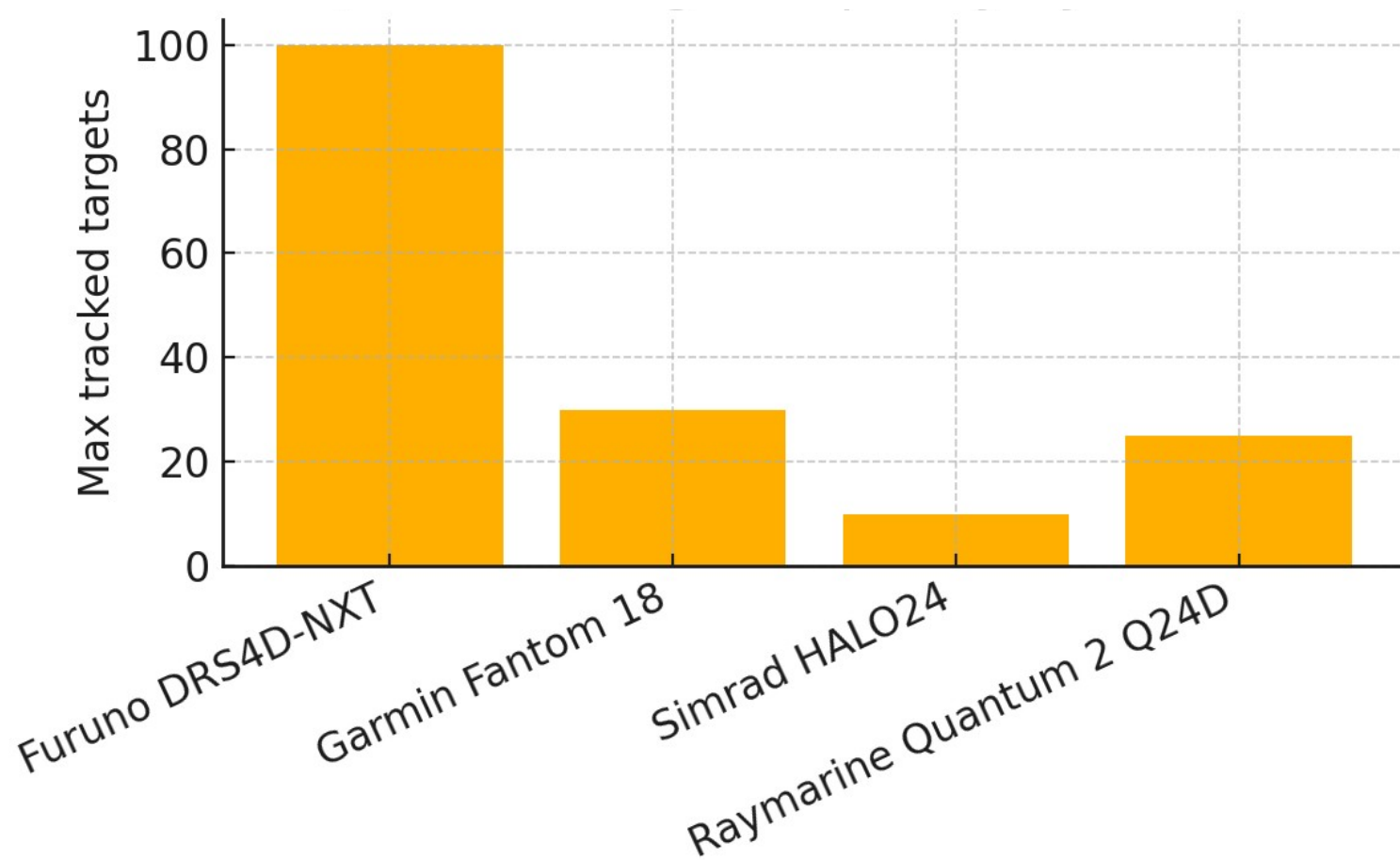
Still valuable for collision avoidance on budget sets.



Garmin Fantom 18: up to 30 MARPA targets, manual acquire only.



Simrad HALO24: up to 10 (20 dual-range).



ARPA/mARPA – Furuno NXT Implementation

- First leisure-class solid-state Doppler radar with auto-ARPA 100 targets.
- Fast Target Tracking in seconds.
- Can detect low-RCS objects, i.e. kayaks and birds.
- Min range ≈ 20 m; max instrumented 12 NM ARPA.
- Single-range (up to 48 RPM).
- Dual-range (2 windows ≤ 12 NM, 24 RPM).
- RezBoost sharpening to 2.0° beam.
- Keeps targets even after changing range settings.

ARPA/mARPA – Radar ARPA Comparison

Radar model	Auto-tracking?	Max targets	Instrumented range	ARPA / mARPA range	Notable extras
Furuno DRS4D-NXT	Yes; Fast Target Tracking (auto + manual)	100 ARPA	36 NM	12 NM ARPA	RezBoost sharpening; Bird Mode
Garmin Fantom 18	No; manual MARPA only	30 MARPA	48 NM	n/a (manual)	60 RPM $\leq$ 1.5 NM
Simrad HALO24	No; manual MARPA*	10 (20 dual-range) MARPA	48 NM	n/a (manual)	60 RPM $\leq$ 1.5 NM; dual-range
Raymarine Quantum 2 Q24D	Yes; auto & manual MARPA	25 MARPA	24 NM	$\leq$ 25 targets	CHIRP pulse-compression; TrueTrails

PART 3: The Project

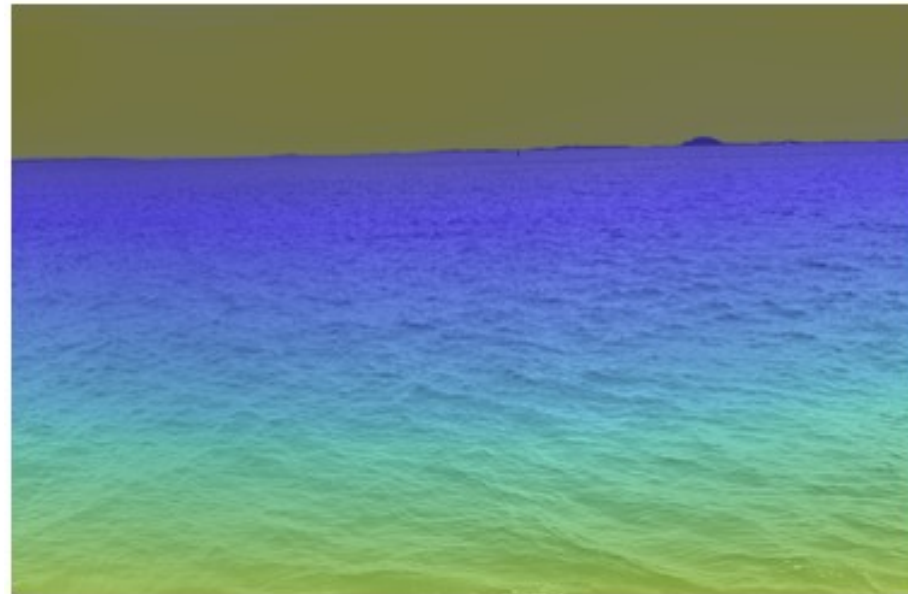
A Succinct Summary



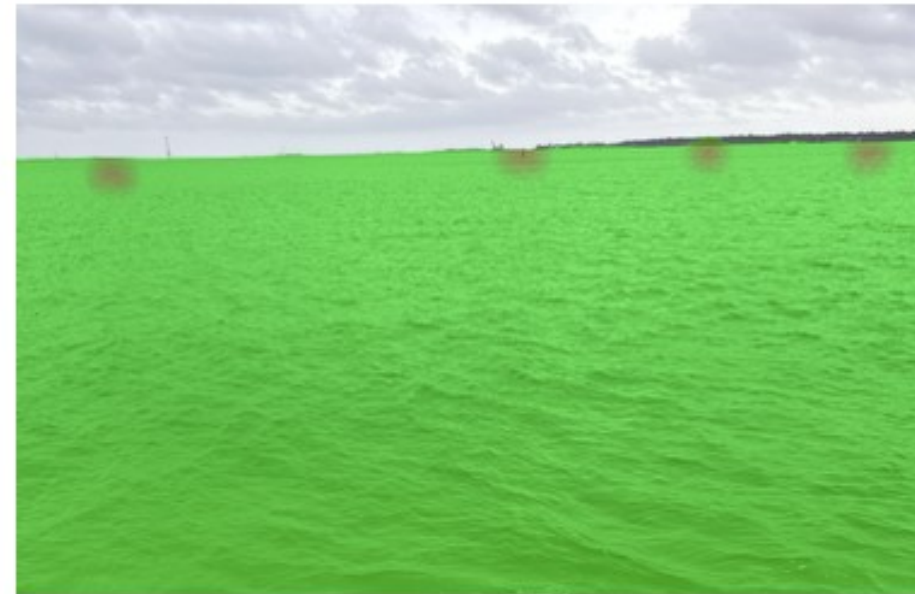
(a) Original image



(b) Segmentation



(c) Depth Map



(d) Radar overlay

Project – An Idea

- Labelling images is time consuming 😞
 - Especially when the objects are tiny in a sea of water.
- Radar could help identify areas of interests for human reviewers OR automated labelling systems



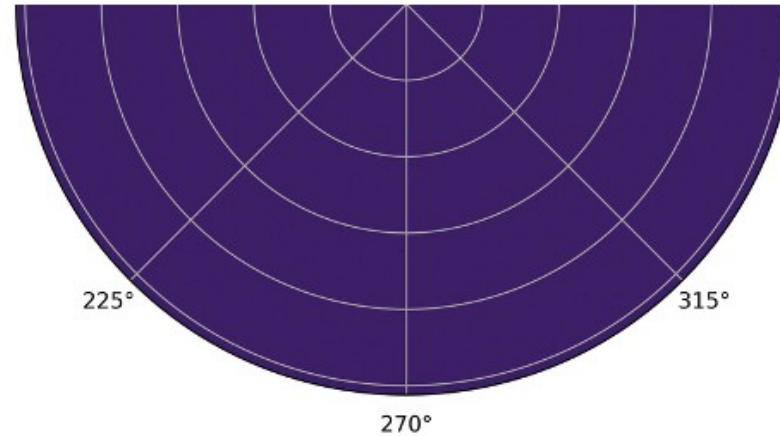
*Not smaller than
32px



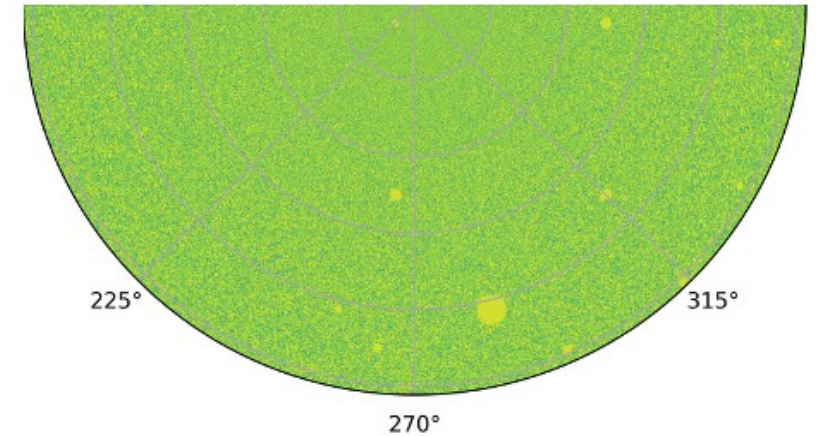
Project – The Problem

- Radar can not always reliably detect an object
 - Especially in high clutter environments
- ... BUT by collecting radar samples at varying gains, one can acquire a more detailed picture

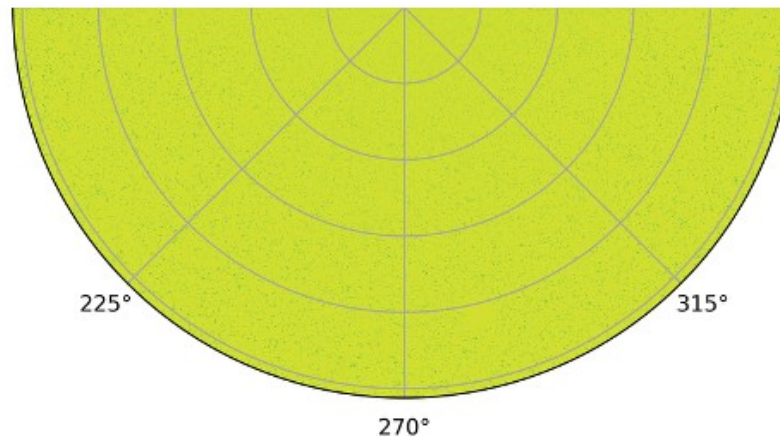
High Clutter Environment



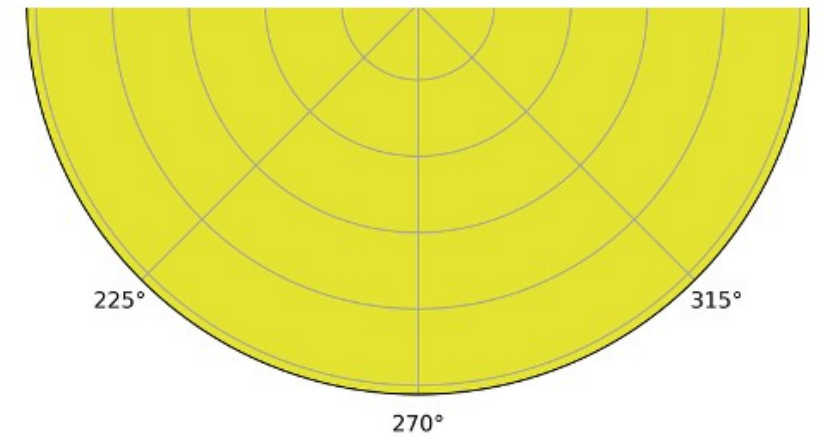
(a) Low Gain



(b) Low-Med Gain

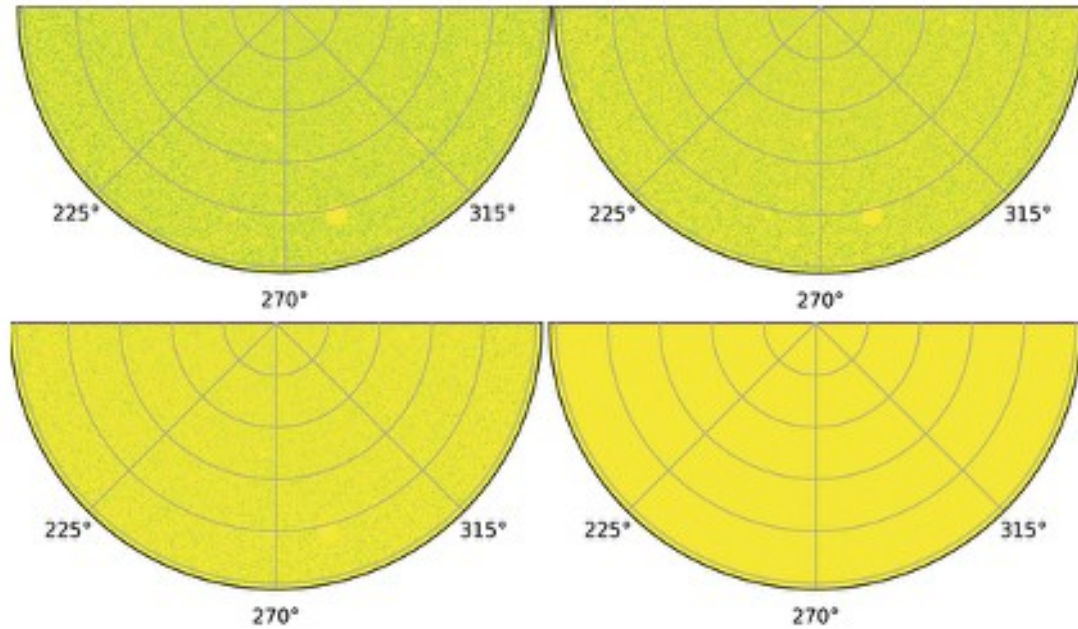


(c) Med-High Gain

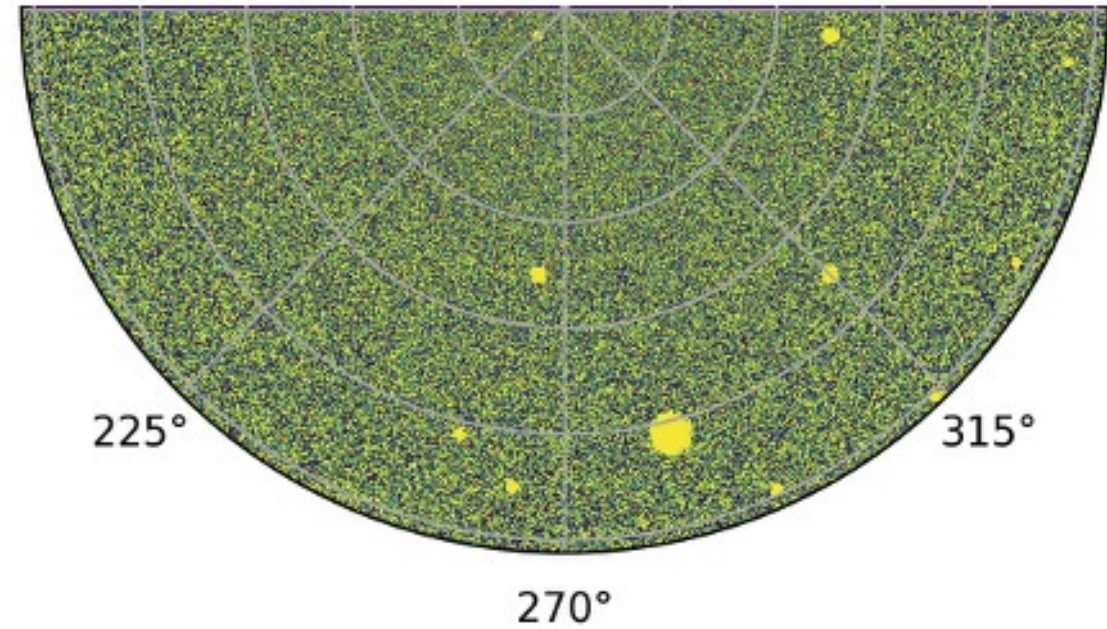


(d) High Gain

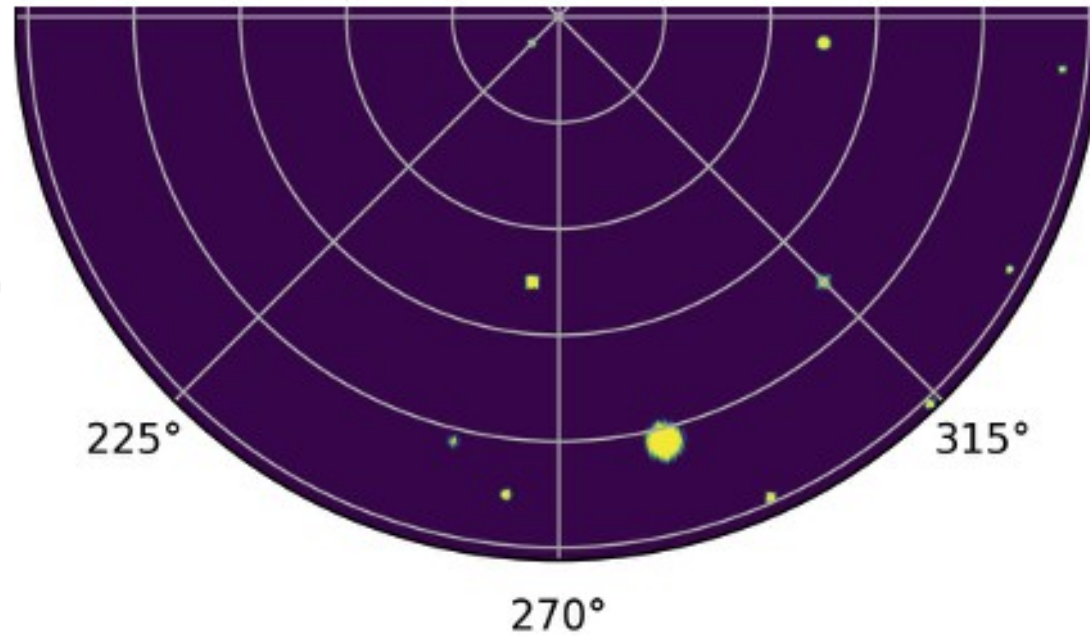
(a) Multi-Gain Capture



(b) ML Clutter Removal



(c) Ground Truth Objects

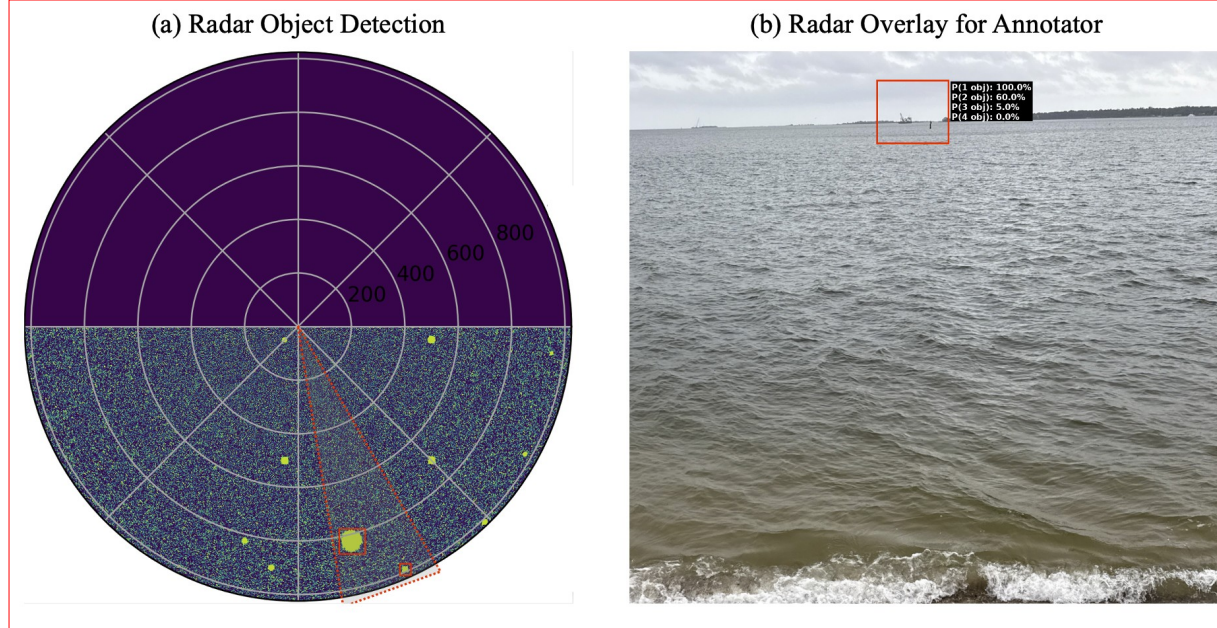


Tossing those frames into a simple algorithm, a clearer picture can be extracted

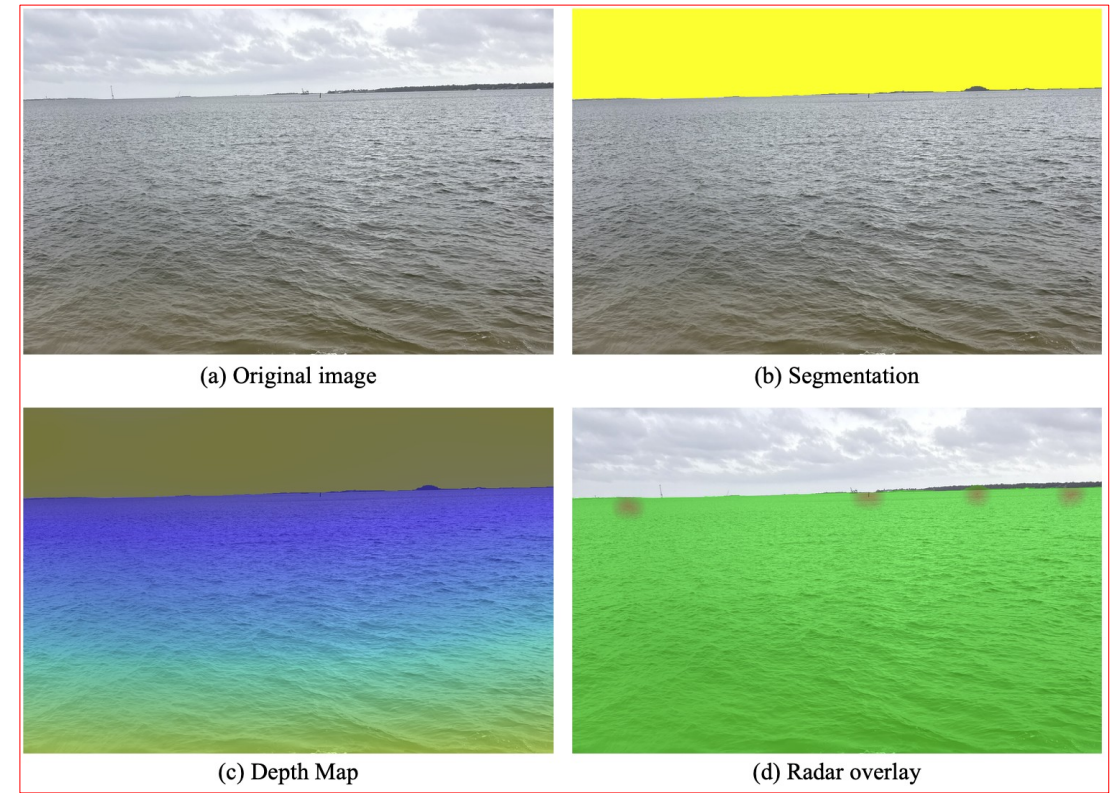
Radar Object Identification Plan

- 3 possible approaches
 - a) 'HDR' the frames together to get better quality images; then use ML to identify objects
 - b) Combine Frames in a way where you can create a statistical gradient across the radar plane and determine likelihood of an object being in a location – non ML
 - c) Develop New ML Architecture to handle Temporal & multi-Gain inputs and detect/classify Objects - HARD(i think)

a)



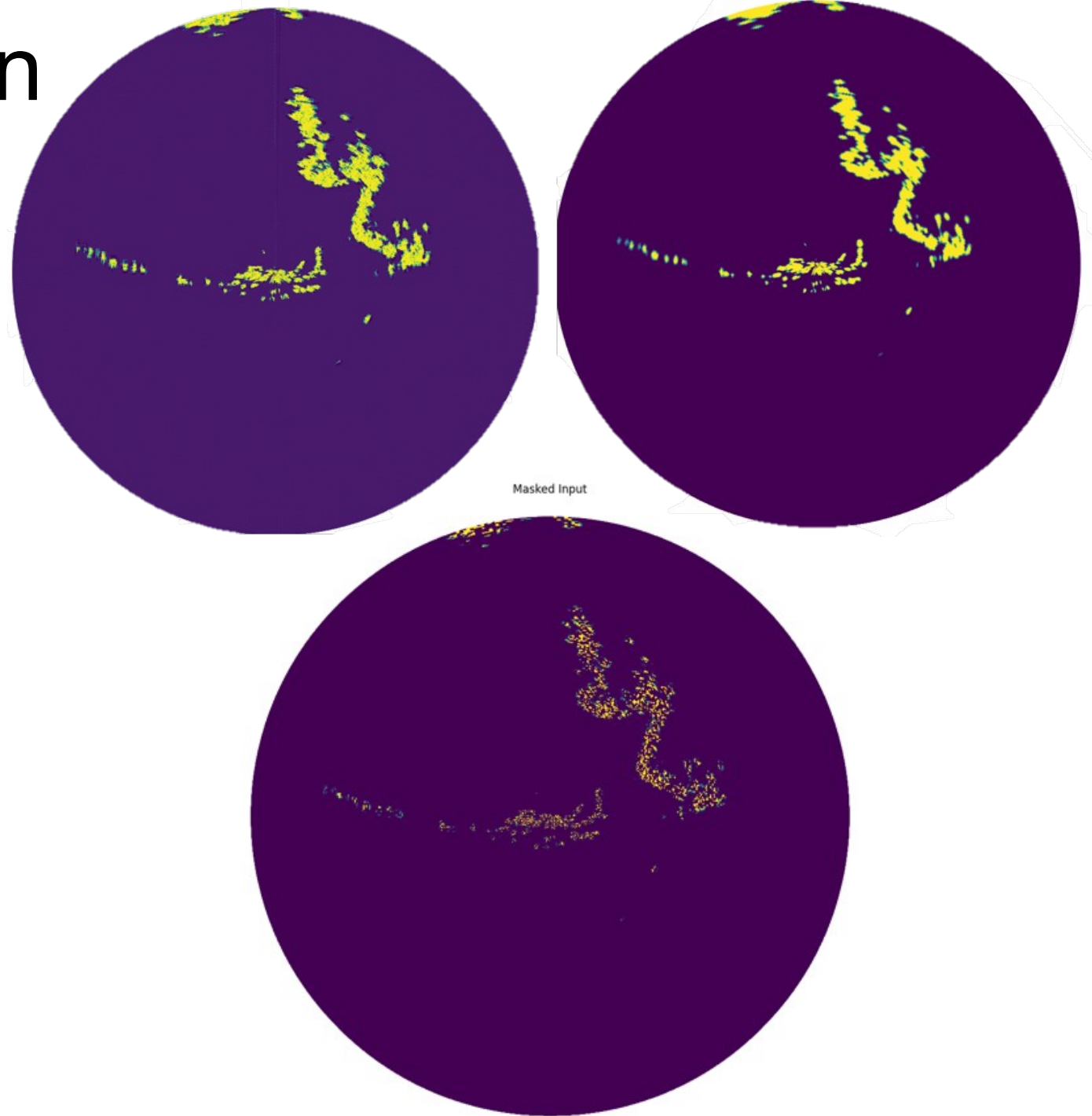
b)



Labelling Reduction

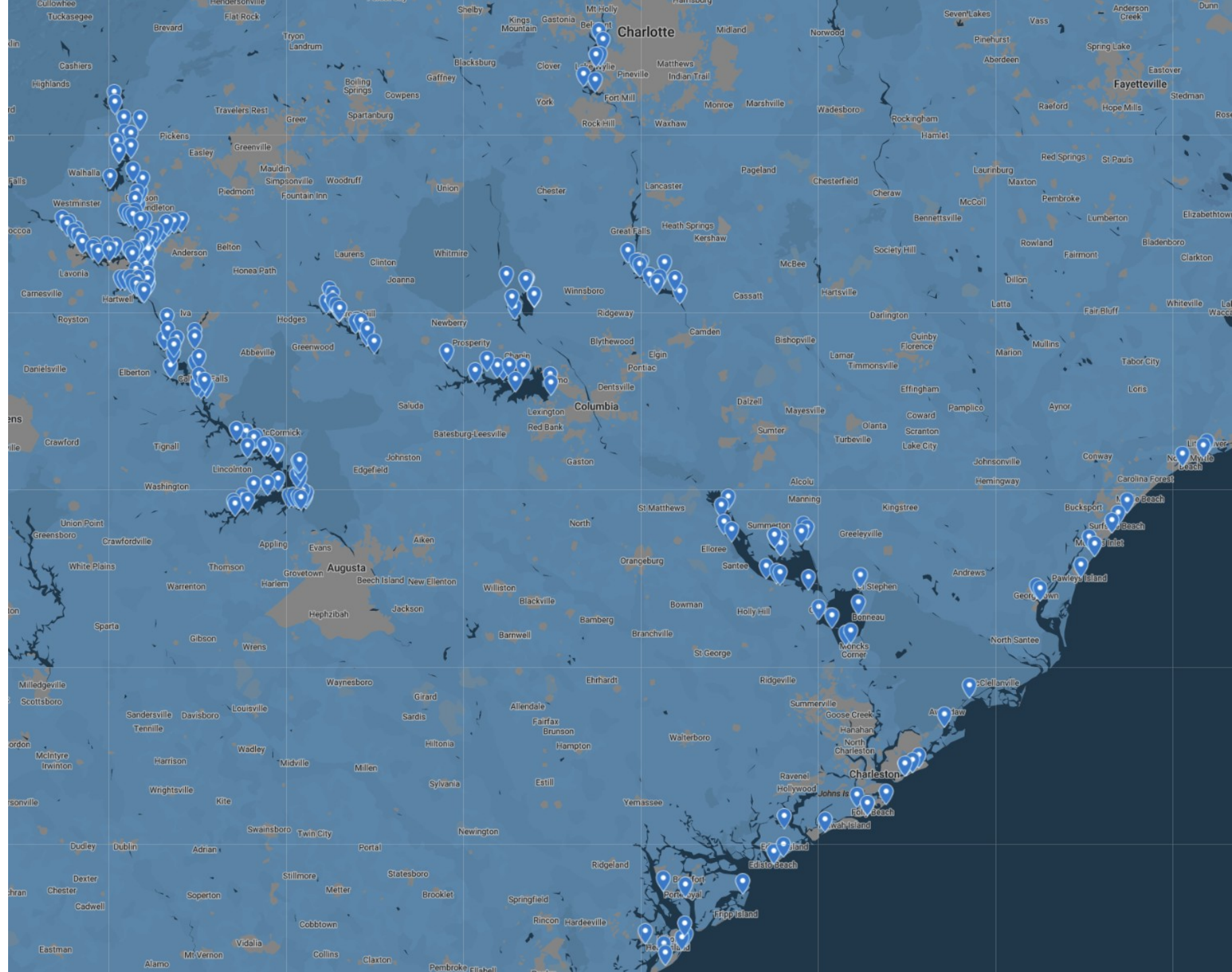
– MAE + detect head

- Self-supervised task
 - No labels, only raw images.
- Feature richness
 - Encoder must infer context, learns shape/texture/semantics.
- Data efficiency downstream
 - Few/one-shot tasks need only a handful of labels once you have the encoder.



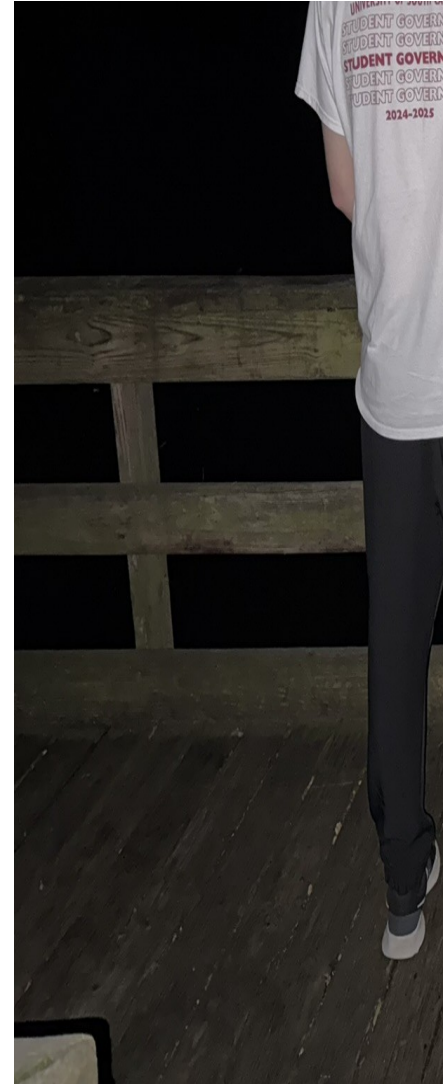
Data Collection – Where?

- With MAE, a diversity of data is good, else the extracted features will be hyper-specific to a certain location.
- I propose these 198 collection points across SC, possibly adding points in VA at some point.



Data Collection – How?

- Collection is primarily Littoral (land-based)
- Future work *should* include open-water collection, but not strictly required.
- Collected with Furuno NXT Radar



Capture Rig Computer Casing

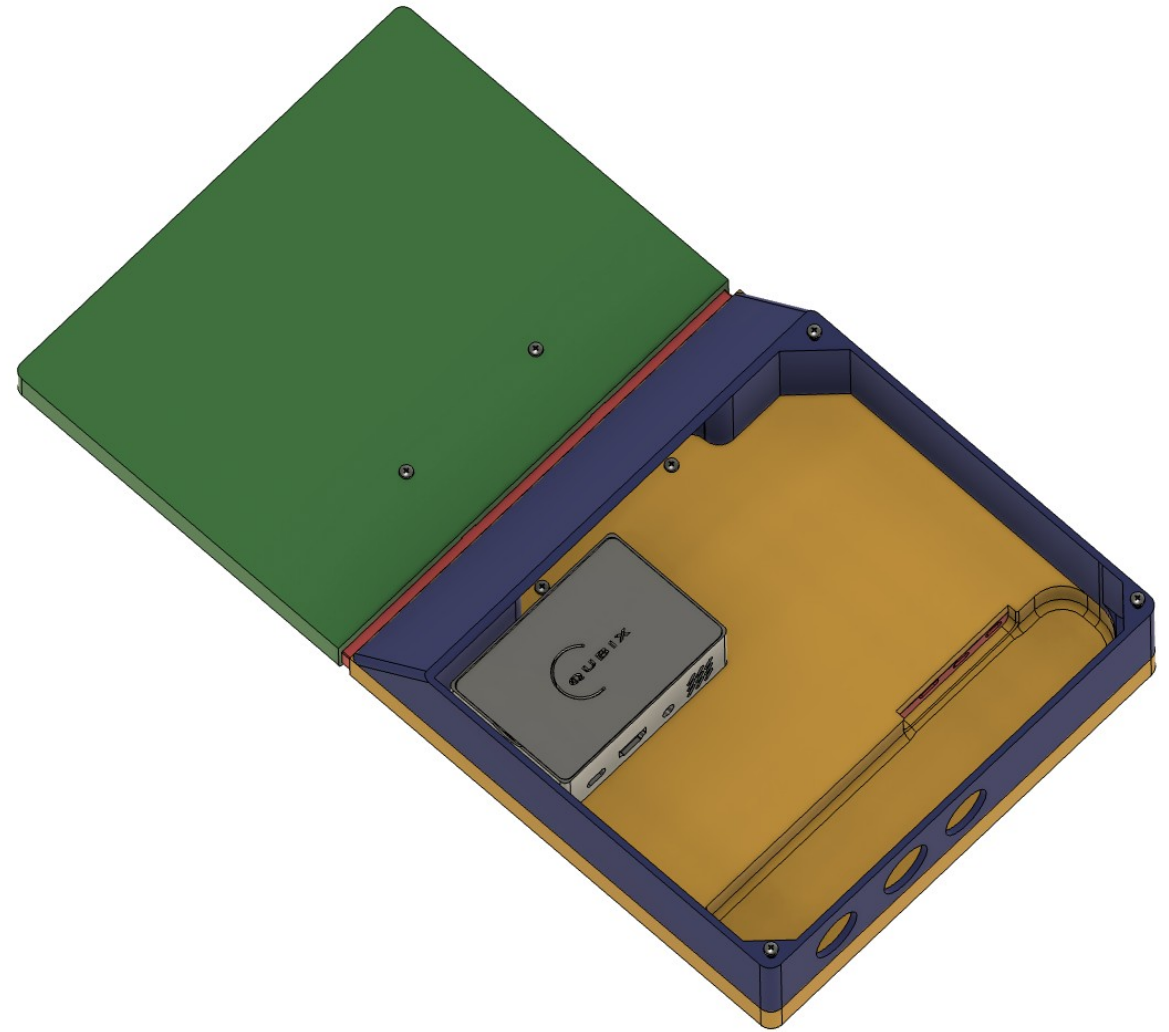
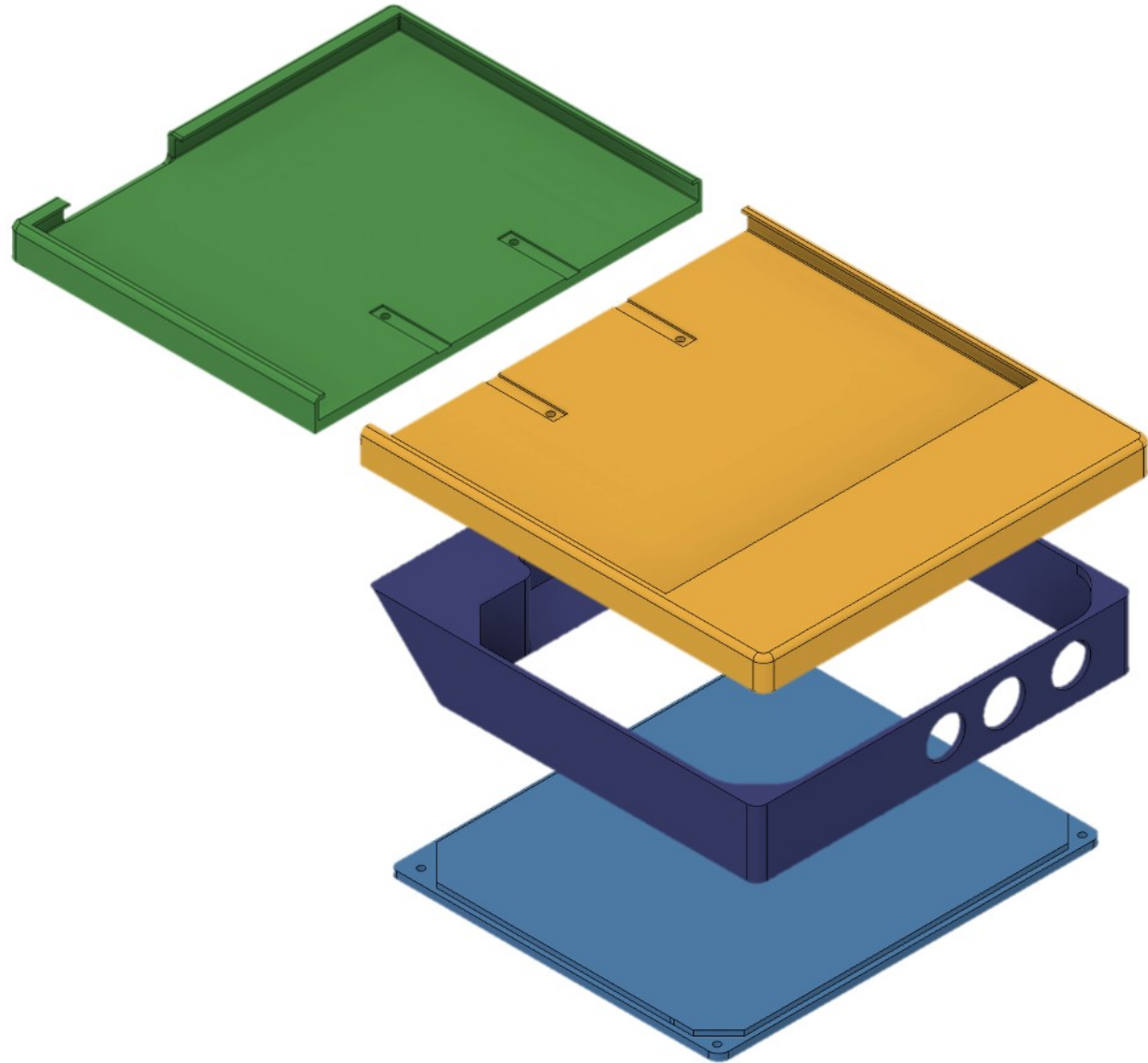


OLD

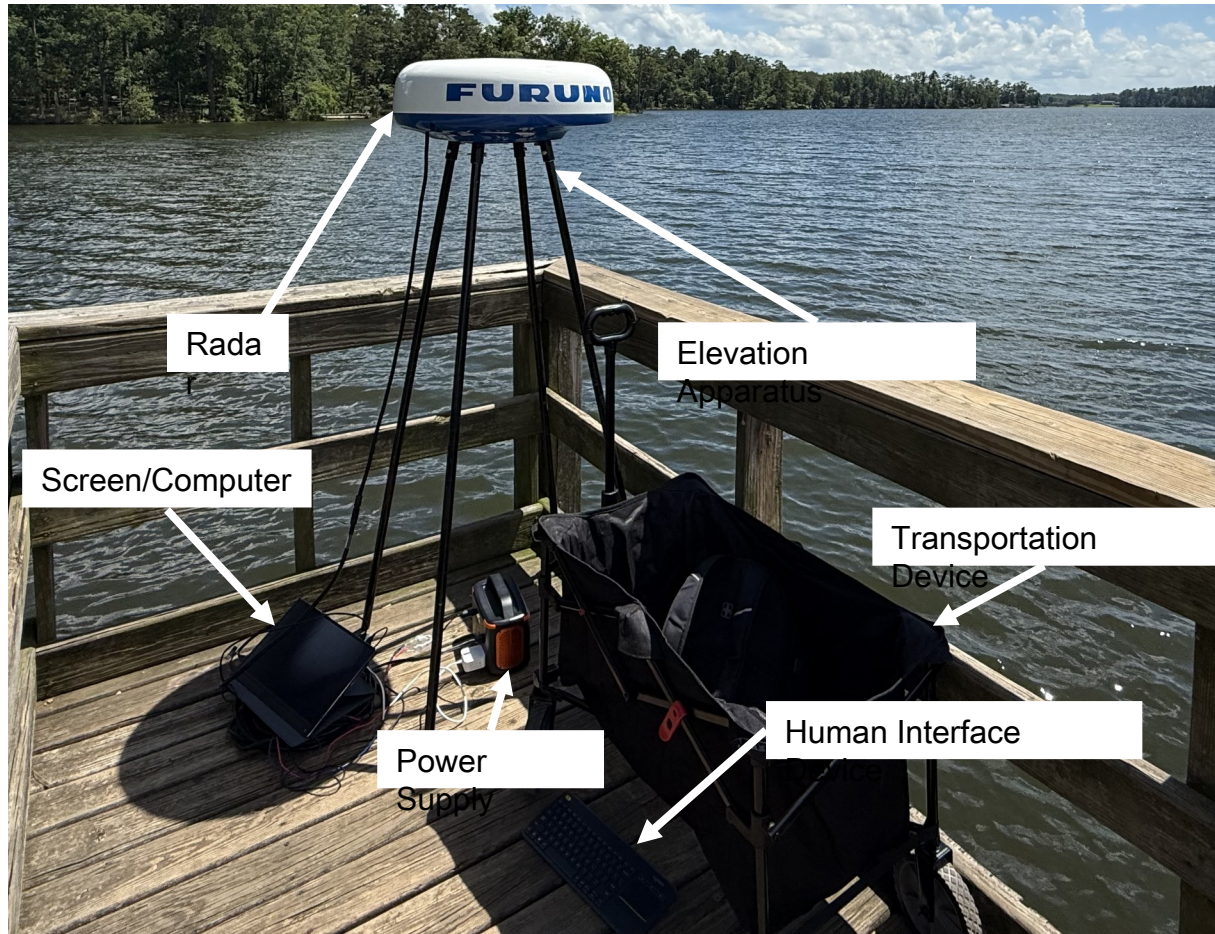


NEW

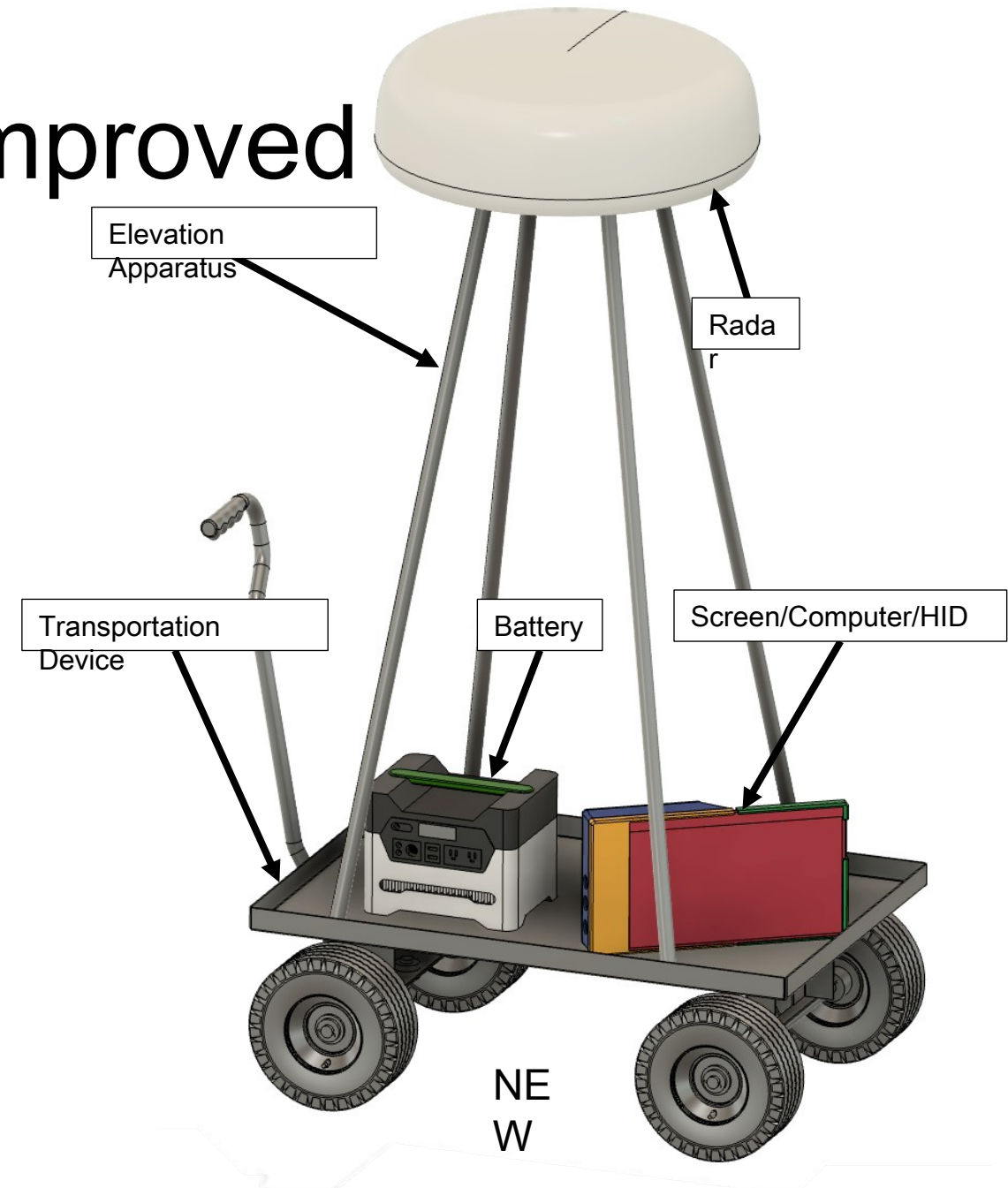
Capture Rig Computer Casing – 3D Printable



Capture Rig – Setup Improved

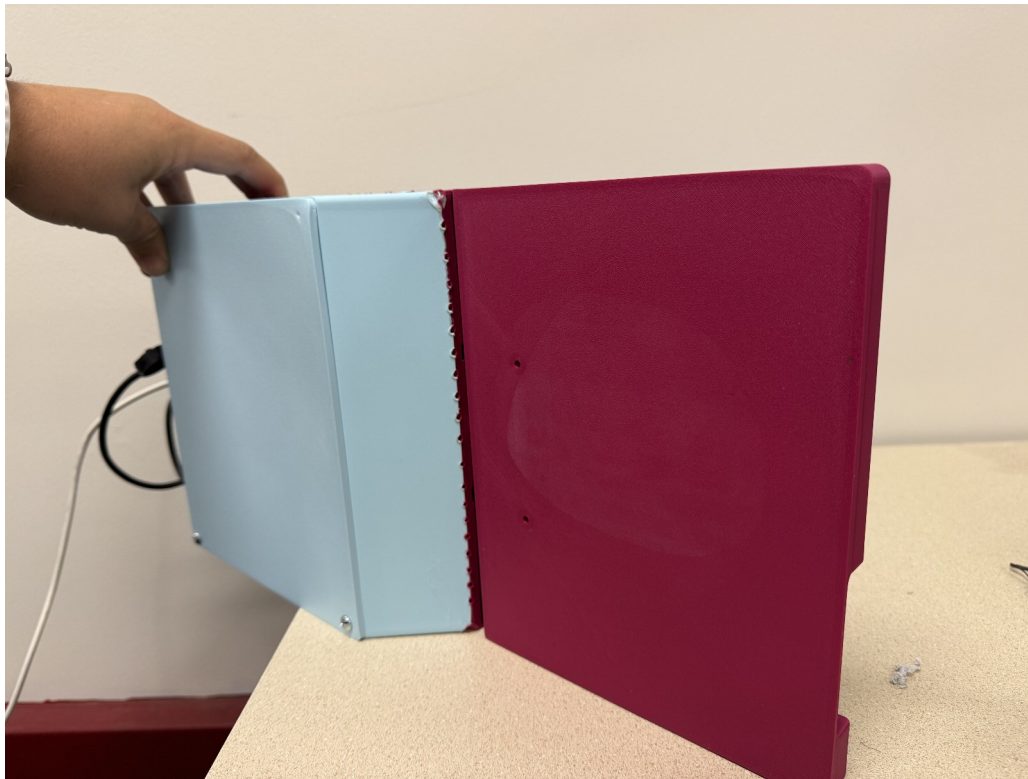


CURRENT



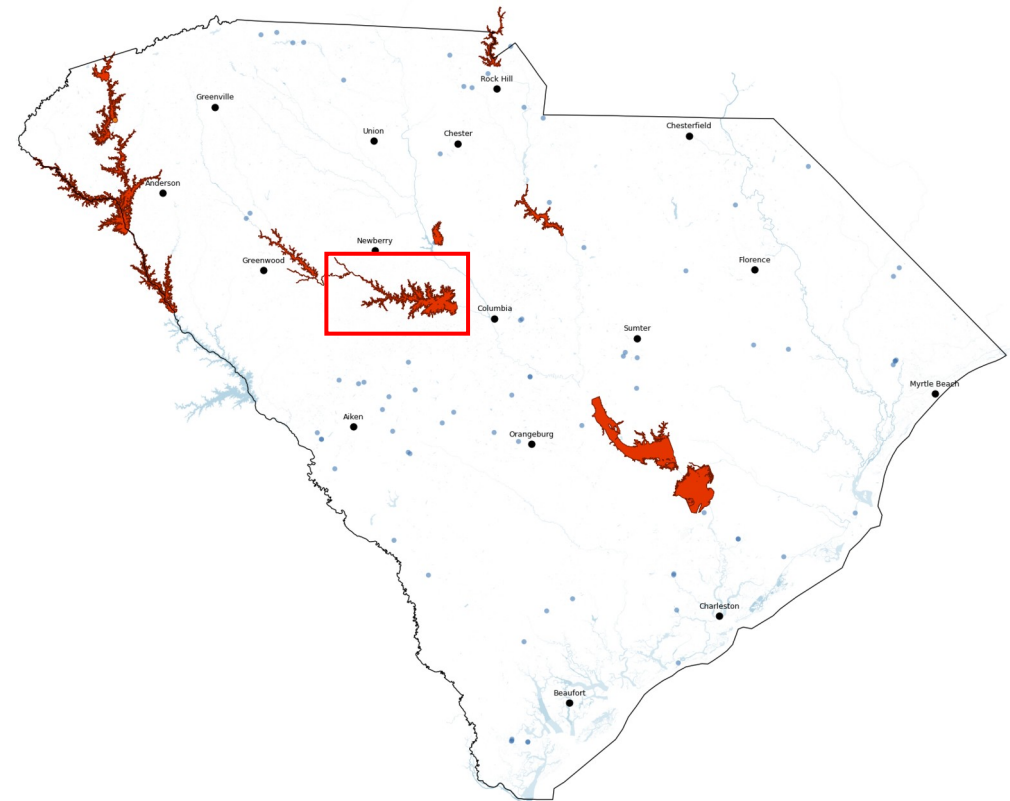
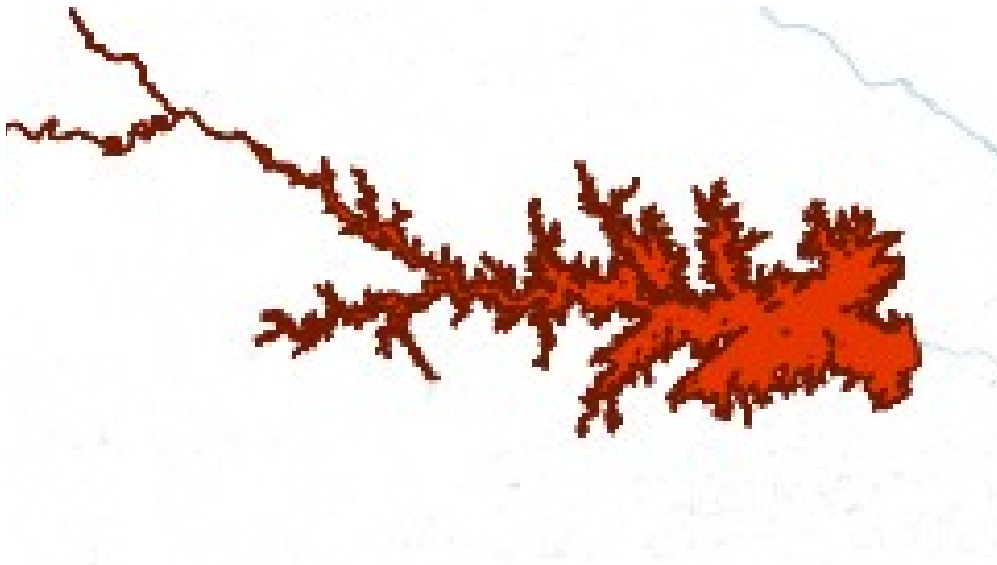
Multimodal Capture Rig

- Will want to add Camera mounting to Radar setup at some point to create finalized system. GPS/IMU would also be useful
 - I think Ty is working on this



Data Collection

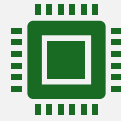
- Collected 50GB(10,000 frames) of data across 6 of Lake Murray's 10 Public Access points as of now.
- Heading, Lat, Lon, and time data are saved.



Data Storage/Processing



Collected data initially in .csv format; very inefficient to store



Use a Rust based script to quickly convert csv to NetCDF4 file, a standard format to access radar data in.



Using NetCDF, I compress just the echo values, without compressing the whole file, making it easy to work with when using MATLAB or python



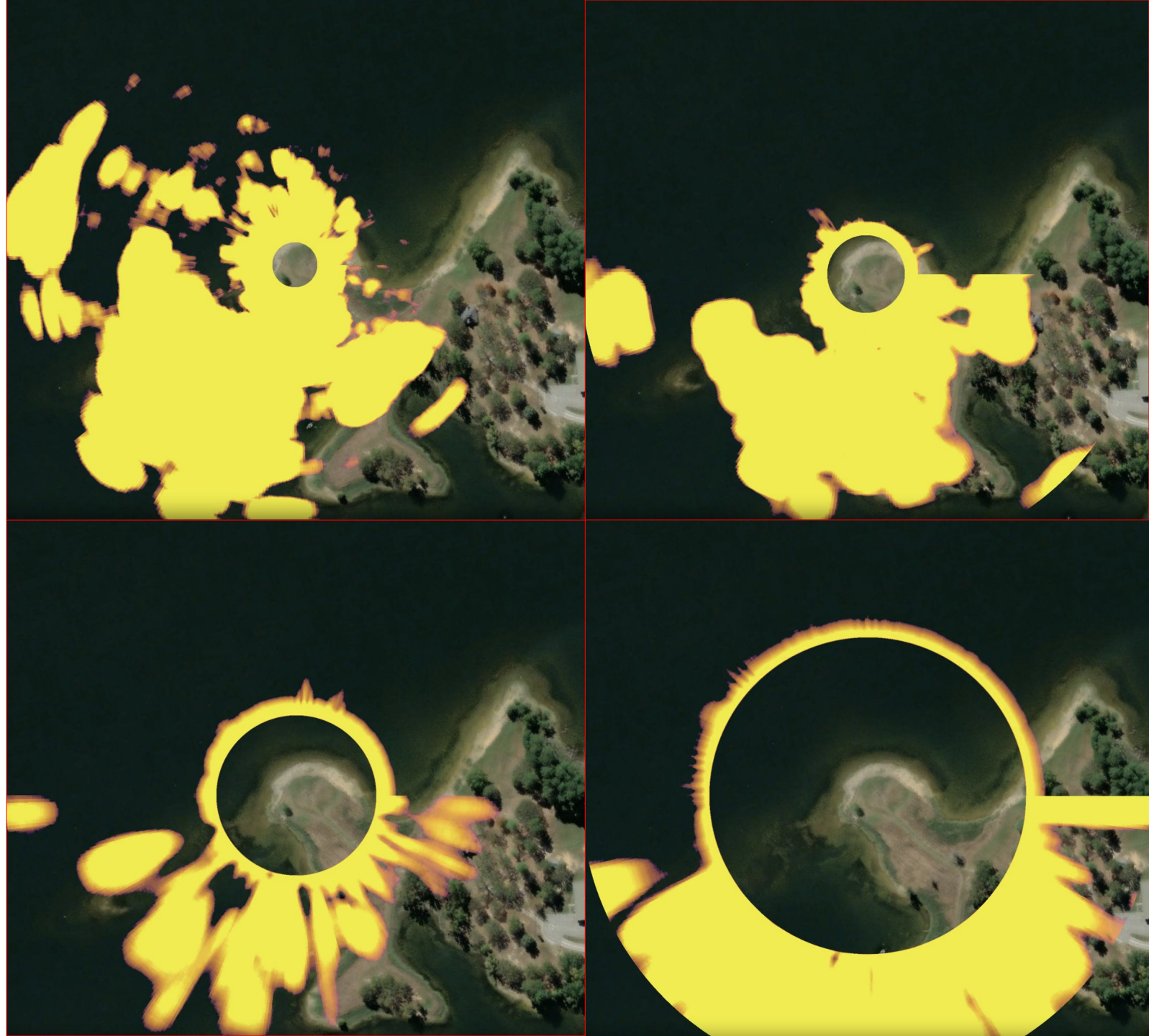
50GB -> less than 2GB -> 25x reduction!

Preview of collected data



Pulse Width

- Tested Pulse Width control



For DRS4D-NXT

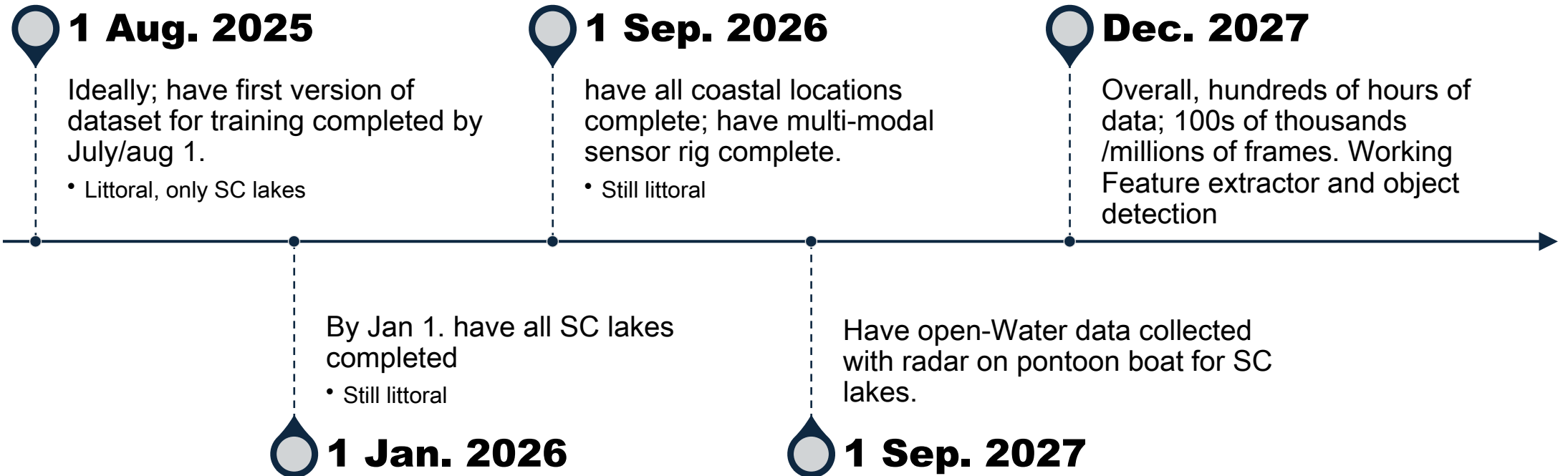
nm/sm	0.0625	0.125	0.25		0.5	0.75	1	1.5	2	3	4
km	0.125	0.25	0.5	0.75	1	1.5	2	3	4	6	8
S1	Normal	Normal	Normal	Normal	Normal						
S2						Normal	Normal				
M1								Normal	Normal		
M2										Normal	Normal
M3											
L											

nm/sm	6	8	12	16	24	32	36				
km	12	16	24	32	36	64					
S1											
S2											
M1											
M2											
M3	Normal	Normal	Normal								
L				Normal	Normal	Normal	Normal				

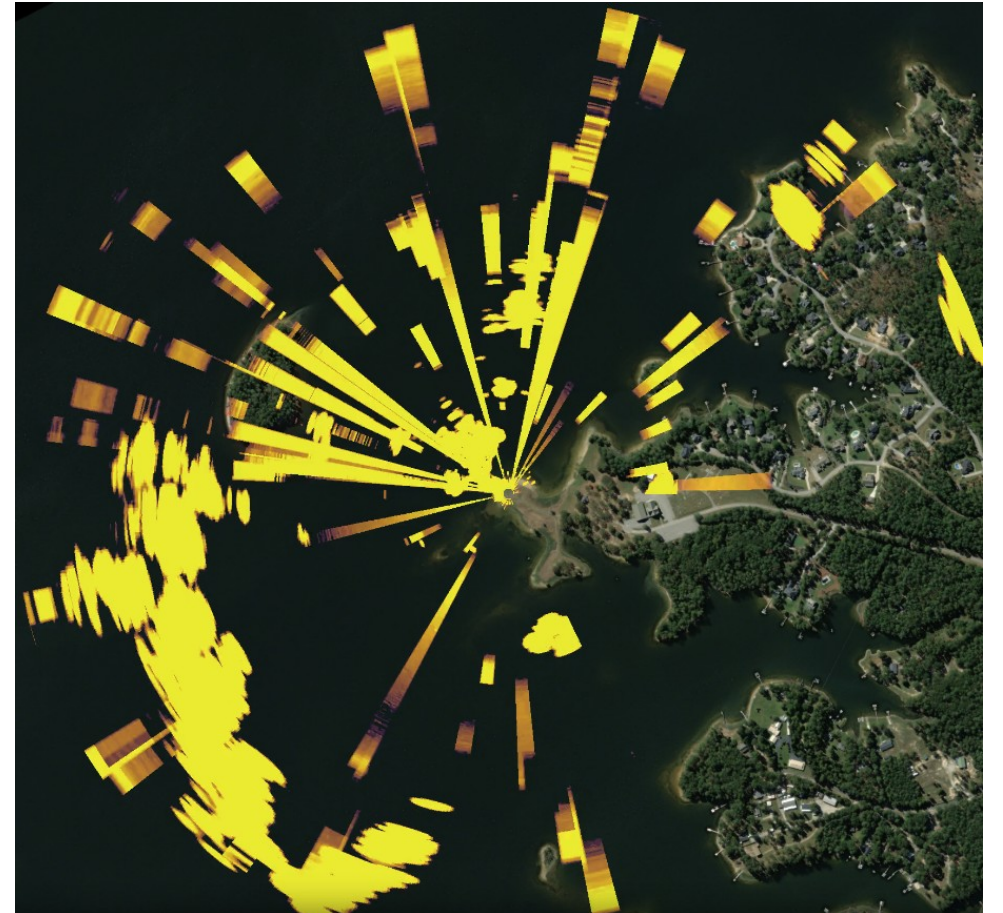
Future Work

- Process more data and figure out if I can fix issues with processing or 'magic'
- Get the testing rig setup time faster (Pi can be slow to boot – 4-6 minutes)
- Add more features to web interface so that things are more intuitive
 - i.e. range values(0.0625 nm) instead of range 'codes'(range 21)
- Running more ML preliminary Tests
- Finding the 'ideal' gain values – currently collect about 60 samples; <10 would be ideal

Timeline



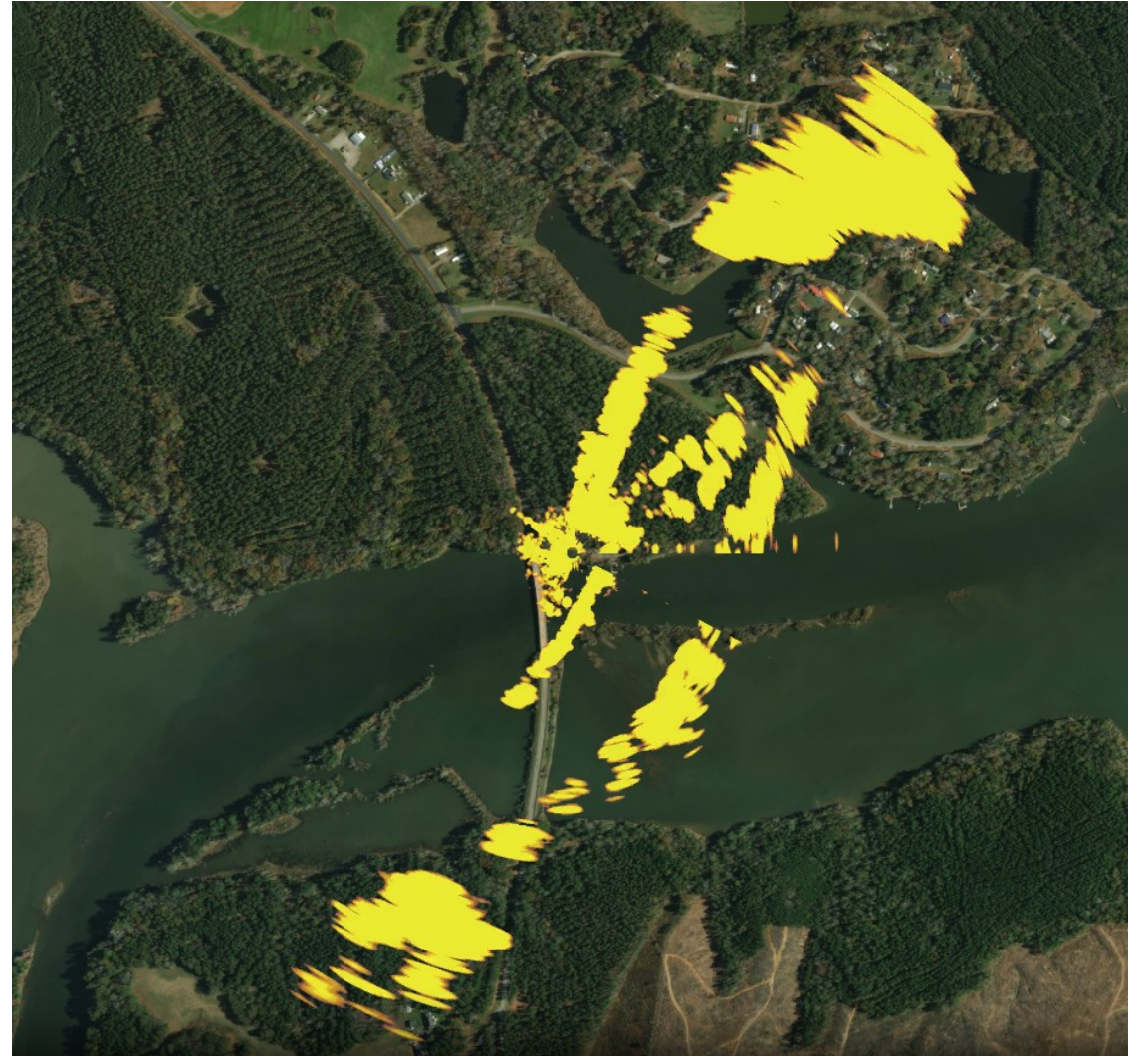
issues



issues



issues



issues

- Missing pulses
 - Getting less than 1000
 - Also less than 1024 echos, but it is consistent, so nothing serious yet
- Pi taking a while to boot, figuring out setup sequence. Need to be better with keeping time accurate(currently some mistakes in time due to sloppy work)
- Not all metadata is collected correctly(i.e. range code, initial params, etc)